Market Size, Product Recalls, and Pharmaceutical Innovation

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Abstract

We examine the causal effect of market size on pharmaceutical research when revenues and innovation are jointly determined. Combining U.S. sales, recall, and clinical-trial data, we use selected major drug recalls as plausibly exogenous instruments for therapeutic-class revenues in a fixed-effects Poisson control-function model. The identifying variation comes from revenue contractions that persist beyond the recall year. A complementary structural demand model examines substitution among medicines and the outside option, helping interpret how product-level disruptions affect class revenues. It compares predicted price responses and substitution under observed recall exposure with a counterfactual that removes the selected recall terms, holding prices and available products fixed. Our preferred elasticity of linked clinical-trial stocks with respect to market sales is 0.79: under the maintained identifying assumptions, a 10% decline in sales corresponds to approximately 8% lower expected trial stock. Separate estimates by clinical phase among studies with documented industry participation are positive and statistically significant for Phase III trials and Phase IV postmarketing research. Estimates for Phases I and II, and those using annual new molecular entity approvals, remain imprecise. These findings indicate that market rewards influence advanced clinical development and research on medicines already available to patients, extending the relevance of innovation incentives beyond the arrival of new drugs.

Keywords: Product recalls; Innovation; Empirical Industrial Organization; Pharmaceutical industry.

JEL Codes: O31, J10, J20, I11, L11

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1 Introduction

That larger markets induce more innovation is a cornerstone of innovation economics (Ayerst, 2025; Klepper and Malerba, 2010; Scherer, 1982; Schmookler, 1966), yet estimating this relationship causally remains elusive when market size and innovation are jointly determined equilibrium outcomes. This identification challenge has concrete policy stakes: the premise that larger expected rewards induce more research activity underlies the design of public demand incentives such as orphan drug legislation (Yin, 2008), making it important to establish whether and how strongly market size drives innovation. Schmookler's demand-pull hypothesis has been challenged on precisely these grounds. As early as the 1990s, Kleinknecht and Verspagen (1990) noted that the direction of causality between market size and innovation was far from obvious, suspecting a simultaneous relationship that their methods could not address. More recently, Stoneman (2010) and Ball et al. (2018) have developed more rigorous approaches to detect this endogeneity, yet a credible causal estimate remains elusive

At the market level, Acemoglu and Linn (2004) proposed a strategy to overcome endogeneity bias by exploiting changes in market size across drug categories driven by demographic trends. After Acemoglu and Linn (2004), the literature shifted from establishing the existence of reverse causality between market size and innovation to identifying the optimal identification strategy. The instrument used in Acemoglu and Linn (2004) was later criticized by Cerda (2007) as being endogenous. As detailed in Cerda (2007), pharmaceutical innovation may increase patient age, while an increase in the average age of patients raises demand for innovative products. Demographic trends may therefore affect market size, which in turn affects innovation, while innovation also influences demographic trends. The importance of such a concern has become central in the AI era, where the possibility of adopting millions of AI agents, each expert in biology, to drive pharmaceutical innovation is increasingly contingent (Amodei, 2024). To the best of our knowledge, the identification strategy for assessing the impact of market size on innovation remains an open issue.

Inspired by Acemoglu and Linn (2004), most authors have examined the relationship between market size and innovation in the pharmaceutical industry (Janssen, 2023), which offers an ideal setting for this question. In this sector, consumer needs are diverse and remain almost stable over time, so the market can be divided into relatively independent submarkets based on medical needs (Bertoni et al., 2010; Sutton, 2001). Moreover, investment in innovation is essential to the industry's existence, and innovation is relatively easy to measure. In the pharmaceutical industry, relevant submarkets are well-defined using the Anatomical Therapeutic Chemical (ATC) classification system, which identifies submarkets based on *"the active ingredients of drugs according to the organ or system on which they act and their therapeutic, pharmacological, and chemical properties"* (World Health Organization, n.d.). Our contribution also considers the pharmaceutical industry as an ideal setting where we provide a novel identification strategy to estimate the causal effect of market size on innovation by exploiting product recalls as plausibly exogenous shocks to expected market rewards.

Therefore, our main contribution is methodological. We employ product recalls as an instrument to address the endogeneity of market size, an issue often overlooked in previous studies. Unlike the traditional Poisson approach, we adopt a Control Function (CF) IV strategy that leverages the unpredictability and abrupt nature of product recalls. The main improvement is the use of recalls as an instrument for market size, directly addressing the endogeneity challenge. This approach relies on the exogeneity of recalls, assuming that firms cannot anticipate competitors' recalls at the market level. The sudden and unexpected nature of major recalls, which are not necessarily more common in "riskier" ATC markets, offers a unique perspective. Our framework links major recalls to clinical research through expected market rewards. A severe product problem can reduce the availability of treatments or confidence in medicines that remain on the market. If substitution towards competing products and price adjustments do not fully offset the lost sales, revenues can decline across the therapeutic class. When this contraction persists, it can lower the expected returns to clinical development. We use observed class revenues as a proxy for these expected rewards and exploit the variation associated with selected major recalls. Market fixed effects account for persistent differences between therapeutic classes, including their baseline recall risk, while year effects and pre-event controls account for common shocks and observed market conditions. Identification requires that, conditional on these controls, the selected recall shocks are unrelated to other unobserved changes in research conditions and affect trial

activity through the market rewards reflected in revenues. In particular, recalls must not independently alter development costs or approval prospects through channels omitted from the model. Under these assumptions, recall-induced revenue changes allow us to distinguish the effect of market rewards on clinical research from the influence of innovation on market size. Section 5.1 discusses these assumptions and examines the evidence supporting their plausibility. The fact that the identifying variation comes from recall-induced contractions in-class revenues are not without consequences. Namely, it assumes that clinical research responds to rising revenues in the same way it responds to falling revenues when applying the estimated elasticity between innovation and market size to market expansions. The constant-elasticity specification imposes this assumption: for small revenue changes, a percentage increase is predicted to raise expected trial stocks by approximately as much as an equally sized percentage decrease would lower them.

Since substitution patterns of products play an important role in defining the strength and the exclusion restriction of our instrument, we complement the market-level analysis with a structural demand model of substitution among medicines and the outside option (Appendix D). The model allows demand to respond to a medicine's own recalls and to recalls affecting other products, and allows own recalls to change price sensitivity. We compare predicted price elasticities and substitution following a small price increase under observed recall exposure with a counterfactual that removes the selected recall terms, holding prices and available products fixed, while allowing for shortages. To the best of our knowledge, this is the first model of this kind on recalls. This exercise gives us hints on the competitive relationships possibly affecting class-level revenue changes, and whose relevance was underlined, but not explored, in Dubois et al. (2015): a sales loss for a recalled product need not translate into an equal loss for its therapeutic market when buyers can switch to other medicines. Consequently, a separate portfolio analysis examines potential sales offsets through other products marketed by the recalling firm (Appendix E).

Furthermore, we extend the analysis to industry-linked research at different clinical phases, including studies conducted after a medicine has reached the market. This allows us to examine the response of postmarketing research, an activity that counts of newly launched drugs do not directly capture.

Our preferred specification yields an elasticity of linked clinical-trial stocks with respect to market sales of approximately 0.79. Under the maintained identifying assumptions, a 10% decline in market sales corresponds to approximately 8% lower expected trial stock.

Finally, we examine the response of clinical-trial stocks separately by phase, focusing on studies with an industry sponsor or collaborator. The aggregate trial count combines research activities that differ in their remaining investment, development risk and time to potential commercial returns. Early trials assess candidates whose commercial prospects remain uncertain, advanced trials are generally closer to potential approval, and Phase IV studies concern medicines already on the market. These differences may affect how firms respond to changes in expected revenues, motivating separate estimates for each phase. The relative responsiveness of these activities is an empirical question.

We find positive and statistically significant elasticities for Phases III and IV, while the estimates for Phases I and II remain not significantly different from zero. The positive estimates are consistent with market rewards influencing advanced clinical testing and post-marketing research, although they do not establish that these activities are more responsive than earlier research. This is quite expected as changes in expected revenues may affect exploratory Phase I trials and subsequent Phase II and Phase III development differently, reflecting differences in remaining costs, development risks, and the time to potential commercialization. The Phase IV result is particularly informative because it concerns research conducted after a medicine has reached the market. It therefore complements evidence based on new-drug launches (Dubois et al., 2015): changes in market rewards may affect research on existing treatments, an activity that counts of newly launched drugs do not directly capture.

The remainder of the paper is structured as follows. Section 2 reviews the related literature. Section 3 presents the theoretical framework. Section 4 describes the data. Section 5 sets out the empirical strategy and examines the identifying assumptions. Section 6 reports the empirical results and Section 7 concludes.

2 Literature Review

The relationship between market opportunities and innovation has long been studied in economics. Schumpeter emphasized the role of large enterprises in technical progress, while the subsequent demand-pull literature examined how demand directs inventive activity. Schmookler (1966) argued that larger markets encourage invention, and Scherer (1982) found positive associations between industrial demand and patenting, together with an important role for technological opportunity. This literature also raised questions about the interpretation of demand–innovation relationships. Mowery and Rosenberg (1979) criticized the broad definitions of demand used in earlier studies and their limited treatment of supply-side influences. Kleinknecht and Verspagen (1990) subsequently questioned the direction of causality and suggested that demand and innovation may be jointly determined. Using UK time-series data, Geroski and Walters (1995) found that economic activity predicted subsequent innovation in a Granger-causality sense. These contributions established the importance of distinguishing the effect of market opportunities on innovation from feedback and other factors affecting both.

This question is particularly relevant to pharmaceutical policy. Prices, insurance coverage and market protection influence access to existing medicines and the returns expected from developing new treatments (Lakdawalla, 2018). Assessing these policies therefore requires evidence on how commercial incentives affect the amount, direction and timing of research. The relevant market encompasses the sales opportunities available to potential developers. Its size differs from that of any individual firm, since a firm can expand by gaining market share even when total demand for treatments in its therapeutic area remains unchanged.

We organize sales, selected recalls and linked trial activity by therapeutic drug class. This common classification allows us to relate disruptions to marketed medicines to revenues and research activity in the same therapeutic area. The economic interpretation of these groups depends on substitution patterns, which motivates our complementary demand analysis in Appendix D. Within these classes, scientific advances, regulation and competition can change over time. Market fixed effects absorb persistent differences across therapeutic areas, leaving these changing influences and reverse causality as identification concerns.

Market rewards and identification. An influential approach uses demographic variation to measure potential demand. The seminal paper of Acemoglu and Linn (2004) combines changes in population and income with fixed age profiles of pharmaceutical expenditure. Their baseline estimates imply that a 1% increase in potential market size induces approximately a 4% increase in the entry of nongeneric drugs and new molecular entities. This approach relates innovation to the population likely to use a treatment, reducing reliance on contemporaneous pharmaceutical sales, which also reflect past innovation.

Demographic measures nevertheless raise a specific feedback question. Cerda (2007) studies how pharmaceutical innovation can reduce mortality and increase the population generating future demand. Larger markets encourage drug development, while new medicines can expand those markets through improved survival. Market size and innovation can therefore reinforce one another over time.
Other papers in the literature rely on demographic shifts as an instrument for market size.

Dubois et al. (2015) use international data to relate the number of new chemical entities reaching the market to reconstructed lifetime revenues, instrumenting revenues with demographic and income variables. Their preferred elasticity is approximately 0.23, relating successful entry to returns over a drug's commercial life. Differences in innovation outcomes, market-size measures, time horizons and identifying variation mean that these elasticities need not measure the same response. The distinction between potential demand and realized revenues matters: revenues depend on prices, competition and available treatments as well as the number of patients. Our analysis uses selected recalls of medicines marketed in the United States to study the response of annual clinical-trial stocks to contractions in annual therapeutic-class revenues. This design adds evidence on research responses to disruptions affecting existing treatments. Its interpretation depends on the conditions discussed in Section 5.1.

Evidence on prices and research also helps explain what revenues represent. Giaccotto et al. (2005) document a positive relationship between pharmaceutical prices and R&D intensity. Civan and Maloney (2009) find that therapeutic categories with higher prices for

existing medicines have more drugs in development. A change in revenues can influence research through expected future returns and, when firms face financing constraints, through the funds available to finance current projects. Blume-Kohout and Sood (2013) discuss both channels in their analysis of Medicare Part D, including the possibility that improved firm valuations reduce the cost of external finance. This distinction also applies to recall-related revenue losses. Our market-level estimates do not separately identify changes in expected profitability and changes in financing conditions, but fixed effects mitigate such concern.

Policy changes provide further sources of variation in commercial incentives. Finkelstein (2004) finds that policies raising expected returns to vaccines stimulated clinical development. Blume-Kohout and Sood (2013) document increased research activity in therapeutic classes more exposed to the expansion of prescription-drug coverage under Medicare Part D. Yin (2008) studies the research response to the incentives introduced by the Orphan Drug Act, and Iizuka and Uchida (2017) find that reduced patient cost sharing for selected rare and intractable diseases in Japan increased firm-sponsored clinical trials. These interventions affect different components of the return to research, including demand, exclusivity and development costs.

The ability to earn returns from a potential market also depends on patent protection and insurers' coverage decisions. Kyle and McGahan (2012) find that stronger patent protection around the implementation of TRIPS increased research associated with wealthier markets, with more limited evidence for diseases prevalent in poorer countries. Agha et al. (2022) show that the introduction of exclusions from pharmacy benefit managers' formularies reduced research in more exposed therapeutic classes, particularly where existing treatments were numerous and development was less scientifically novel. Their results provide evidence on contractions in commercial opportunities.

Research outcomes and development decisions. The outcome used to measure innovation determines which part of the research response is observed. R&D expenditure measures resources committed to research, trial starts measure new studies, and approvals or launches measure products that have completed development and regulatory review. Trial stocks also depend on the duration of recorded study activity. For example, Lichtenberg (2007) studies lengthy cancer-related innovation using counts of chemotherapy regimens and scientific publications on drug therapy, finding positive associations between both measures and disease incidence.

Pammolli et al. (2011) link declining pharmaceutical R&D productivity to changes in the risk and complexity of the projects pursued. Research inputs and successful products can therefore evolve differently. Scientific opportunities also matter: Rake (2017) finds that growth in relevant scientific knowledge helps explain pharmaceutical innovation alongside potential market size. This evidence motivates accounting for changes in scientific activity in our empirical specification.

Development data already feature prominently in the literature, including Kyle and McGahan (2012) and Cunningham et al. (2021). They allow researchers to observe projects that continue, fail or reach the market. The scientific content of those projects can also respond to demand. Dranove et al. (2022) find that the response to Medicare Part D was disproportionately concentrated in follow-on research. Consequently, a larger research response need not imply a proportionately larger increase in scientifically novel or therapeutically valuable treatments. We interpret linked trial stocks as a measure of recorded research activity. Within the linked sample, studies are included without requiring them to lead to a newly approved medicine. The outcome therefore covers activity that may never appear in counts of new-drug approvals. Differences between their elasticity and elasticities based on new-drug launches can reflect study entry, duration, attrition and the time required to obtain approval.

The timing of commercial rewards affects development decisions. Budish et al. (2015) show how long commercialization lags can discourage private investment in cancer research, particularly when those lags reduce effective patent protection. Gaessler and Wagner (2022) study patent invalidations and find that their effects on subsequent development and commercialization are largely driven by invalidations early in development. Blume-Kohout and Sood (2013) find a substantial Phase I response to Medicare Part D and a later response in Phase III. These findings motivate allowing responses to vary across clinical phases without assuming that later phases must be more responsive. Our phase-specific estimates concern

studies with documented industry sponsorship or collaboration, separating activities with different remaining costs, risks and times to potential commercialization.

Research also continues after initial market entry. Dubois et al. (2026) examine firms' decisions to seek approval for supplemental uses of existing drugs. They find that relaxing restrictions on truthful off-label promotion reduced the approval hazard for supplemental uses relative to original indications. Their results show that commercial incentives can affect the development of medicines already available to patients. Our Phase IV analysis examines a related margin: the response of postmarketing trial activity to therapeutic-class revenues. It extends the outcomes considered beyond research preceding the first launch of a new chemical entity.

Expected returns also depend on rival development decisions. Rao (2020) estimates a dynamic model using Phase III data in which firms respond to competitors' research choices. Khmelnitskaya (2023) distinguishes clinical failure from withdrawal for commercial reasons, allowing continuation decisions to depend on rival projects. Cunningham et al. (2021) show that acquired drug projects are less likely to continue when they overlap with the acquirer's product portfolio, especially under weaker competition. This evidence helps explain why changes in clinical activity reflect strategic continuation decisions as well as scientific progress. Recalls may alter the competitive conditions under which these decisions are made.

Recalls, substitution and research incentives. Adverse events affecting marketed medicines have already been used to study research responses. Krieger et al. (2022) examine FDA public health advisories and find that affected firms increase research expenditure through greater reliance on acquiring external innovations, particularly when their existing pipelines are weak. Competing developers move resources away from the affected therapeutic areas. Thus, the research response of the affected firm can differ from that of the therapeutic market. Our class-level trial outcomes measure activity within the affected market.

The revenue effect of an adverse event depends on buyers' information and available alternatives. Jovanovic (2021) models recalls as signals about quality and firms' effort, with consequences for reputation, prices and firm value. Freedman et al. (2012) document sales losses following toy recalls that extend to manufacturers whose products were not recalled. In pharmaceuticals, Higgins et al. (2021) examine the demand and valuation effects of safety-related relabeling. Lost sales can move to competing medicines or other products supplied by the same corporation; adverse information may also reduce demand beyond the recalled product. A product-level sales loss therefore need not produce an equal contraction in therapeutic-class revenues.

Structural demand studies provide methods for examining these adjustments. Dubois and Lasio (2018) study pharmaceutical demand and pricing under regulatory constraints, while Dubois and Tunçel (2021) estimate physicians' prescribing responses to safety information and recommendations about antidepressants. Outside pharmaceuticals, Bachmann et al. (2023) distinguish adverse reputation spillovers from substitution gains following the Volkswagen emissions scandal. Our demand exercise in Appendix D compares predicted price elasticities and diversion following a small price increase under observed recall exposure and after removing the selected recall terms, holding prices and available products fixed. Appendix E complements these conditional predictions with descriptive evidence on potential sales offsets within the recalling corporation. Both exercises help interpret the revenue changes underlying the market-level first stage, which includes competitors' sales.

Adverse information may also affect research directly. Dranove and Olsen (1994) find that stock-market responses to dangerous-drug announcements before the 1962 FDA amendments were consistent with anticipated increases in testing costs. Such channels matter for the exclusion restriction: a recall could alter expected research costs, approval prospects or firm-specific financing conditions independently of the class-level revenue change. Our identification argument requires the selected recall variation to be unrelated to omitted determinants of research and to affect trial activity through the market rewards represented by class revenues, conditional on the specified controls and fixed effects. Section 5.1 develops this argument and examines evidence relevant to its plausibility.

Our central contribution is therefore evidence on the response of pharmaceutical research to revenue contractions associated with selected recalls. We estimate this response using a fixed-effects Poisson control-function framework following Lin and Wooldridge (2019). The phase analysis extends the investigation to industry-linked postmarketing research. The

demand and portfolio analyses examine substitution and potential sales offsets that help determine how a disruption to an individual medicine translates into changes in market revenues.

3 Theoretical Framework

This section outlines the conceptual framework underlying our identification strategy. We formalize how product recalls affect expected market rewards and, in turn, firms' innovation decisions. In our model, innovation choices are forward-looking and respond to the discounted stream of expected market rewards, not merely contemporaneous sales.[1] Furthermore, the observed market size is measured in terms of revenue and is therefore an equilibrium outcome that may change with both demand and supply. The goal is to formalize the conditions under which a major recall causes a persistent shift in expected market rewards at the therapeutic-class level, while remaining consistent with the exclusion restriction embedded in our control-function IV Poisson specification. To do this, we will draw in part on Jovanovic (2021). Product recalls can change the set of available medicines and the valuation of products that remain available. These changes redistribute purchases among competing medicines and the outside option. We partially represent this substitution through the demand system presented in Appendix D, which provides a complementary analysis for products that experience partial or temporary shortages. The CES formulation below provides an illustrative class-level representation of this mechanism, while the appendix uses a random-coefficients discrete-choice model to estimate product-level substitution patterns. Class revenue is the sum of revenues across products, so its response to a recall depends on substitution, the prices of the products purchased, and any subsequent price adjustments. The innovation analysis uses observed class revenue as a proxy for expected market rewards. Identifying the innovation response requires additional assumptions concerning the recall instrument and the relationship between observed revenues and expected rewards, which we discuss below. Consider an ATC-3 therapeutic class indexed by m and discrete time t. Let $\mathcal{J}_{mt}$ be the set of products available in class m at time t. Each product $j \in \mathcal{J}_{mt}$ delivers an effectiveness/quality level $a_{jt} > 0$. To capture imperfect substitution across products within a therapeutic class, define an aggregate "treatment service" through a CES aggregator[2],

$$\mathcal{A}_{mt} \equiv \left(\sum_{j \in \mathcal{J}_{mt}} a_{jt}^{\sigma-1} \right)^{\frac{1}{\sigma-1}}, \qquad \sigma > 1, \tag{1}$$

where σ governs within-class substitutability (hence, σ will be higher in ATC-4 markets), and Equation 1 summarizes the variety-adjusted effective quality in the class. Demand for the class-level treatment service is assumed proportional to $\mathcal{A}_{mt}$,

$$Q_{mt} = D_{mt}\,\mathcal{A}_{mt}, \tag{2}$$

where $D_{mt} > 0$ is a demand fundamental capturing prevalence and diagnosis intensity, prescribing norms and guidelines, reimbursement generosity, and (importantly in this context) perceived safety risk and reputational considerations at the class level [3]. Class-level sales are

[1] Notice that although R&D decisions are forward-looking, our empirical specification is a reduced-form intensity equation in which current innovation responds to the current state of expected market rewards. The model does not require state dependence in innovation counts beyond what is captured by market fixed effects, year effects, and the control function; adding lagged innovation would instead address serial correlation in the unobservables, which we handle through fixed effects and clustered inference.

[2] We introduce a CES aggregator over products within an ATC-3 class as a parsimonious way to capture imperfect substitution among therapeutically related drugs. In pharmaceutical markets, substitution is typically high but not perfect due to contraindications, adherence, clinical guidelines, and switching costs; hence the removal of a major product need not be fully offset by competitors, implying a class-level contraction in quantity and revenues. The CES form is not essential to our results; any aggregator with finite substitution yields the same qualitative implication: product removal can reduce class-level effective variety/quality and thus market rewards.

[3] The term D_{mt} summarizes observed and unobserved class-level demand conditions. We do not estimate its components separately. These conditions can affect both revenues and innovation incentives, indicating another channel through which observed market size may be endogenous. Identification, therefore, relies on the stated assumptions regarding the selected recall variation, rather than on observing all demand fundamentals. In particular, after accounting for the specified controls and fixed effects, recalls must not affect innovation through omitted channels outside the market-reward mechanism represented in the empirical equation.

revenues,

$$\text{Sales}_{mt} = P_{mt}\, Q_{mt}, \tag{3}$$

where P_{mt} is a class-year price index. At annual frequency in U.S. pharmaceutical markets, the class-level price index P_{mt} is largely mediated by institutional features—most directly, administered reimbursement benchmarks that rely on lagged average sales prices in Medicare Part B (Acquatella et al., 2026)—and by contracting frictions in coverage and rebates (e.g., Part D/PBM arrangements) (Berndt and Newhouse, 2012; Duggan and Scott Morton, 2011). Although these institutional features shape pricing incentives, they do not by themselves determine the revenue response to a recall. We do not model P_{mt} explicitly. A recall-induced quantity contraction lowers revenues if changes in the class-level price index—including changes associated with the mix of medicines purchased—are insufficient to offset it, so that $\Delta \log P_{mt} < -\Delta \log Q_{mt}$. An exact offset, $\Delta \log P_{mt} = -\Delta \log Q_{mt}$, would leave class revenues unchanged. Whether recalls generate a net revenue contraction is therefore assessed empirically through the first-stage estimates.

Taking logs and combining (2)–(3) yields

$$M_{mt} \equiv \log(\text{Sales}_{mt}) = \log P_{mt} + \log D_{mt} + \log \mathcal{A}_{mt}, \tag{4}$$

which motivates our interpretation of M_{mt} as a reduced-form proxy for the market rewards relevant to innovation incentives. Innovation is measured by N_{mt}, the number of active (activated) clinical trials in class m and year t. A maybe clearer word than "active" for trials would be "pursued" trials. Activation, indeed, as we will see, is an economic decision which implies entering the trials procedure (Phase I, Phase II and so on). "Activated" refers to trials that are already at a certain stage of the process in that year. For them, the decision is to continue the trial procedure (i.e., keep investing in the medicine to conduct trials in subsequent phases). In the market-size tradition (e.g., Acemoglu and Linn, 2004; Dubois et al., 2015), innovative activity is increasing in the expected present value of profits from a marginal project, which scales with expected market rewards. We therefore define V_{mt} as the incremental expected present value of activating one additional (representative) innovative project targeted to class m at time t:

$$V_{mt} \equiv \mathbb{E}_t \left[\sum_{\tau \geq 0} \beta^\tau \, \pi(\text{Sales}_{m,t+\tau}) \right], \qquad \beta \in (0,1), \tag{5}$$

where $\pi(\cdot)$ denotes the per-period operating profits attributable to that project and is increasing in market revenue (through demand, coverage, and appropriability). Under standard regularity conditions (in particular, π increasing), higher expected revenues $\{\text{Sales}_{m,t+\tau}\}_{\tau \geq 0}$ imply a higher project value V_{mt}, formalizing the forward-looking nature of trial activation decisions.

To connect V_{mt} to the observed count N_{mt}, we adopt a standard "many potential projects" framework for innovation counts (Dranove et al., 2014; Dubois et al., 2015; Hausman et al., 1984). Fix an ATC-3 market m and year t. Suppose there is a large pool of potential trial opportunities indexed by k. A project is activated when its net expected value is non-negative:

$$\text{activate}_{k,mt} = 1\Big\{ V_{mt} - \chi_{mt} + \varepsilon_{k,mt} \geq 0 \Big\},$$

where χ_{mt} captures market-year costs and constraints (regulatory burden, congestion in recruitment, scientific difficulty, etc.) and $\varepsilon_{k,mt}$ is an i.i.d. idiosyncratic component across projects. Under the common type-I extreme value assumption for $\varepsilon_{k,mt}$, the activation probability is increasing in V_{mt} and proportional to $\exp(V_{mt} - \chi_{mt})$. With many independent opportunities, the number of activated projects in (m,t) is well approximated by a Poisson count with intensity proportional to that activation rate, so that $N_{mt} \mid \mathcal{I}_{mt} \sim \text{Poisson}(\lambda_{mt})$ with $\lambda_{mt} \propto \exp(V_{mt} - \chi_{mt})$.

In the data, we do not observe V_{mt} (nor firms' information set) directly. Instead, we observe a revenue-based measure of market size $M_{mt} \equiv \log(\text{Sales}_{mt})$, which in (4) is a reduced-form function of demand fundamentals and effective variety/quality, and is therefore a natural empirical proxy for the profitability conditions that enter V_{mt} at annual frequency (Acemoglu and Linn, 2004; Dubois et al., 2015). Let S_{mt} denote the latent state of expected market rewards relevant to firms' current innovation decisions. We assume that the value of an additional representative project is increasing in this state,

$$V_{mt} = v(S_{mt}), \qquad v'(\cdot) > 0.$$

Expressing S_{mt} in units of log revenue, we represent observed market size as a noisy measure of this state:

$$M_{mt} = S_{mt} + \nu_{mt},$$

where ν_{mt} captures the discrepancy between current revenues and the state of expected market rewards, including measurement error and transitory movements in revenues.

This representation allows current revenues to provide information about expected rewards without requiring them to capture every determinant of project value. Identification, however, requires additional restrictions on the selected recall variation. Conditional on the specified controls and fixed effects, recalls must be unrelated to the unobserved components of expected project value and development costs left outside the revenue specification introduced below. They must also not affect innovation through those omitted components. The identification subsection discusses these assumptions and the empirical checks used to assess specific threats to their plausibility.

Each ATC-3 market m is subject to a certain latent defect intensity at time t, denoted by d_{mt}, capturing severe product-level safety or manufacturing problems that may trigger regulatory intervention in the style of Jovanovic (2021). The latter is represented by $d_{mt} = \bar{d}_m + \eta_{mt}$, where $\bar{d}_m$ is a time-invariant component reflecting the baseline risk profile of the therapeutic class and η_{mt} is an idiosyncratic severe-defect shock. Modeling the full FDA decision process is beyond the scope of the present paper. Nevertheless, it is reasonable to assume that the regulator acts to protect patients' health and therefore withdraws products from the market when sufficiently severe adverse effects are detected. In this sense, a major recall can be viewed as the observable realization of a latent defect event once its seriousness crosses a regulatory threshold. Formally, one may write

$$R_{mt} = \mathbf{1}\{d_{mt} > \bar{d}\},$$

where R_{mt} denotes a major recall event and $\bar{d}$ is the threshold above which the defect becomes severe enough to trigger regulatory action [4]. In the spirit of Jovanovic (2021), d_{mt} cannot be predicted by either the regulator or the firm, and $\bar{d}$ is private information of the regulator. Furthermore, in our context, a major recall is interpreted as a public signal of a severe product-level problem that depresses expected market rewards at the ATC-3 level. We therefore summarize the effect of recalls on the latent profitability state as

$$S_{mt} = \bar{S}_{mt} - \psi_0 R_{m,t} - \psi_1 R_{m,t-1} - \psi_2 R_{m,t-2} + \xi^s_{mt}, \qquad \psi_0, \psi_1 > 0,$$

where $\bar{S}_{mt}$ is the underlying profitability state absent recall events, and ξ_{mt} captures other shocks to expected market rewards. The R_{mt} and $R_{m,t-1}$ terms capture the immediate contraction in expected revenues induced by the recall, while the doubly lagged term allows for gradual recovery.

Let κ_{mt} denote an unobserved profitability component (e.g., scientific opportunity, guideline shifts) that affects trial activation through net costs/constraints and may be persistent over time [5]. If κ_{mt} is persistent, then M_{mt} and its lags can remain correlated with κ_{mt} through the dependence of S_{mt} on persistent unobservables, so replacing M_{mt} with $M_{m,t-1}$ does not generally deliver causal identification. In other words, lagged market size may improve timing or forecasting, but it removes endogeneity only under the strong assumption that the unobservables generating endogeneity have no relevant temporal dynamics (Bellemare et al., 2017). This motivates instrumenting M_{mt} in our empirical analysis. Basically, although trial activation is forward-looking, it is based on information available at time t: because the lag from clinical development to marketing is long and uncertainty about distant payoffs is substantial and heavily discounted (DiMasi et al., 2016; Glied and Smith, 2013; Keyhani et al., 2006), firms effectively condition on the current profitability state of the market. Empirically, contemporaneous market size M_{mt} provides a noisy but informative signal of this persistent state, whereas lagging M_{mt} does not in general remove endogeneity

[4] More generally, the model works even if one wants to model recalls as an intensity rather than a binary event. In this case, R_{mt} can be interpreted as arising from a stochastic process with arrival rate increasing in d_{mt}

[5] Here ξ^s_{mt} captures shocks to the latent profitability state, whereas κ_{mt} below denotes the reduced-form unobserved component entering the empirical innovation equation.

when unobserved profitability determinants are persistent.

In the data, we do observe contemporaneous market size $M_{mt} = \log(\text{Sales}_{mt})$ and controls z_{mt}, together with market fixed effects and year effects. Absorbing the proportionality constant and time-invariant heterogeneity into c_m, and writing the market-year cost shifter as $z'_{mt}\gamma + \kappa_{mt}$, yields the intensity specification that matches our empirical framework:

$$\mathbb{E}[N_{mt} \mid M_{mt}, z_{mt}, c_m, \kappa_{mt}] = c_m \exp\Big(\beta_1 M_{mt} + z'_{mt}\gamma_c + \kappa_{mt}\Big). \tag{6}$$

Here $\gamma_c := -\gamma$, and c_m captures time-invariant ATC-3 heterogeneity (absorbed by ATC-3 fixed effects), while κ_{mt} is an idiosyncratic profitability component potentially correlated with M_{mt}, which motivates instrumenting M_{mt} using major recalls within our control-function IV Poisson estimator. This implies that major recalls affect innovation incentives indirectly, by shifting the latent profitability state S_{mt}, which in turn is reflected in the observed revenue-based measure of market size M_{mt}.
Formally Equation 6 is recovered as follows. First we better specify the number of activated projects as a Poisson count with intensity proportional to the activation rate: $N_{mt} \mid \mathcal{I}_{mt} \sim \text{Poisson}(\lambda_{mt}), \quad \lambda_{mt} = \Lambda_m \exp(V_{mt} - \chi_{mt})$, where we added a time-invariant scale $\Lambda_m > 0$ (the size of the opportunity set) in market m.
Second, because S_{mt} is not observed and M_{mt} is its empirical proxy, in the spirit of Acemoglu and Linn (2004) we adopt the following reduced-form approximation:

$$V_{mt} = \alpha_m + \beta_1 M_{mt} + u_{mt},$$

where α_m captures time-invariant ATC-3 heterogeneity and u_{mt} is an idiosyncratic profitability component. Similarly, decompose activation costs as $\chi_{mt} = b_m + z'_{mt}\gamma + \omega_{mt}$, where b_m is time-invariant, z_{mt} are observed cost shifters/controls, and ω_{mt} is an unobserved market-year cost shock. Substituting these decompositions into the intensity yields

$$\lambda_{mt} = \Lambda_m \exp\Big((\alpha_m - b_m) + \beta_1 M_{mt} + z'_{mt}\gamma + (u_{mt} - \omega_{mt})\Big).$$

Define $c_m \equiv \Lambda_m \exp(\alpha_m - b_m)$ and $\kappa_{mt} \equiv u_{mt} - \omega_{mt}$. Then the conditional mean of the Poisson count satisfies Equation 6 which is the mean function estimated in our fixed-effects Poisson framework.
Note that product quality a_{jt} enters the analysis only through the effective-quality aggregator $\mathcal{A}_{mt}$ and therefore through revenues $M_{mt} = \log P_{mt} + \log D_{mt} + \log \mathcal{A}_{mt}$; in the reduced-form approximation below, this channel is summarized by M_{mt} rather than modeled separately.

4 Available data and data construction

This study combines four main data sources. Sales data are drawn from the Evaluate Pharma dataset, which provides U.S. pharmaceutical revenues at the product and molecule levels from 2004 to 2015, subsequently aggregated to the ATC-3 level. Data on clinical trials and Investigational New Drugs (INDs) for the same period are sourced from ClinicalTrials.gov and a Pharmaceutical Industry Database (PHID), respectively.[6] Recall data are manually collected from multiple sources, including the FDA website, openFDA, Onakpoya et al. (2016), the WHOCC website, PubMed, and Siramshetty et al. (2016). Control variables are derived from Evaluate, PHID, and the FDA's website. Some of the variables – such as competition variables and whether a drug used a molecule that was innovative relative to a firm, as well as firm age– are manually constructed, while others – patent expiration, product batch, generic status, and others – are already provided by the data. We have also collected the adverse events (AERS) dataset provided in openFDA.

Our measure of market size is the logarithm of aggregated sales at the ATC-3 level, calculated as the total revenue of products competing within each therapeutic class in a given year, adjusted to 2015 dollars using the Consumer Price Index. The ATC classification system categorizes drugs into five levels, with higher levels indicating greater detail.
We decided to adopt the ATC-3 and ATC-4 classifications, rather than the broader classifications commonly used in the literature, firstly because the ATC-3 level more precisely reflects the structure of submarkets. Traditionally, the literature has focused predominantly

[6] PHID is a dataset maintained at IMT Lucca. For more information, see Pammolli et al. (2011).

on ATC-1 and ATC-2 levels, with ATC-3 and ATC-4 serving mainly as a basis for comparison with these broader classifications. Importantly, adopting the ATC-3 level in our analysis is motivated not only by its finer granularity but also by its alignment with the categorizations used by antitrust agencies. This alignment underscores the relevance of ATC-3 for deriving policy implications and offers a framework that mirrors the operational parameters considered in regulatory assessments.

We construct the controls separately for ATC-3 and ATC-4 classifications. Annual market sales sum quarterly retail and hospital revenues in the original price basis. Product counts distinguish international product names, while firm counts distinguish corporations. Concentration is measured by the Herfindahl–Hirschman index (HHI), calculated as the sum of squared corporations' revenue shares. Generic share is the fraction of products with at least one generic-type presentation sold in that year. Product age uses recorded launch information, with observed-sales fallbacks. We also build firm age. We average these ages within each class and square the resulting means.

Scientific activity is measured using PubMed journal articles indexed to the class's ingredients. Ingredient identities are matched to official NLM MeSH chemical records, supplemented by verified RxNorm aliases. Each article counts once per class-year, using its journal-issue publication year. We use a fixed basket of ingredients marketed during 2004–2015 and retain only classes with complete mapping and retrieval. The scientific-activity controls are

$$g_{m,t-2} = \log(1 + P_{m,t-2}) - \log(1 + P_{m,t-3}), \qquad g^2_{m,t-2},$$

where P denotes the article count. The transformation accommodates years with no indexed articles.[7].

In what follows, we provide more details on the adopted instrumental variable, recalls, and the dependent variable, i.e., the innovation measure.

4.1 Data on recalls

Drug recalls involve the removal or correction of marketed products because of identified defects or health risks. They may be initiated by manufacturers or undertaken at the FDA's request. The FDA distinguishes three risk classes: Class I concerns potentially serious health consequences or death; Class II concerns temporary or medically reversible consequences, or a remote probability of serious harm; Class III concerns products unlikely to cause adverse health consequences FDA U.S. Food & Drug (2019). Notice that these classifications describe health risks, not the size or duration of the resulting economic disruption. A Class I recall may concern particular batches or formulations and need not remove all competing versions of a medicine.

We distinguish these recalls from documented withdrawals of medicines for clinical safety or efficacy reasons. Such withdrawals may concern an entire medicine or a particular marketed formulation and are not necessarily preceded by a Class I recall.[8]

The collected catalog is broader than the set of candidate instruments. For our application, the instrument must predict changes in therapeutic-class revenues relevant to the expected market rewards that influence innovation (Acemoglu and Linn, 2004). A recall affecting particular batches or one supplier may cause only a temporary interruption (e.g., typically Class II recalls), with competing medicines absorbing the displaced sales. Evidence on prescribing responses to safety information illustrates how substitution can accompany reductions in the use of affected treatments (Dubois and Tunçel, 2021). Consequently, a

[7]MeSH indexing can encompass related substances, and articles need not concern the ingredient's specific therapeutic indication. Current indexing is applied retrospectively. Medicine matches, exact queries and source records are retained in the replication materials.

[8]Here, "withdrawal" denotes a documented safety- or efficacy-related removal from sale. This differs from the FDA's technical category of "market withdrawal," which concerns minor violations not warranting legal action, or situations involving no violation.

Our dataset combines FDA recall records with separately verified clinical safety or efficacy actions. We retain records that can be linked to a product and its recorded corporation in the sales data, excluding unresolved matches, uncertain product scope, and purely commercial discontinuations. For consistency across market definitions, retained products must have an unambiguous assignment at both ATC3 and ATC4.
The reader is referred to Appendix B for a detailed description of how the recalls have been manually collected.

decline in one product's sales need not imply a contraction of its therapeutic market (even when the recall is classified as Class I). Moreover, clinical safety news may alter regulatory scrutiny or development decisions independently of market revenues, threatening the exclusion restriction.

We therefore use medical severity as an initial screen for potentially substantial disruptions, focusing on Class I recalls alongside separately verified clinical safety or efficacy actions. As mentioned, our interest is in actions that may affect revenues across a therapeutic class, given the importance of potential market rewards for pharmaceutical innovation (Acemoglu and Linn, 2004). This motivates focusing on documented failures that compromise the safe and reliable delivery of an already marketed treatment, potentially affecting both its availability and confidence in its use. When such failures interrupt usable supply, competing suppliers may be unable to replace the affected medicines immediately because of production and regulatory frictions, even in generic markets (Conti and Wosińska, 2025). Liebman et al. (2023) provide a concrete example by documenting the reduced vaccination following a vaccine shortage lasting eighteen months that originated in sterility concerns.

The consequences of a recall may also extend beyond the products removed from sale if concerns about their safety affect confidence in related products. Freedman et al. (2012) document recall-related sales losses among manufacturers whose products were not recalled, illustrating this possibility.[9] These availability and confidence mechanisms provide the economic rationale for our selection criterion. The three definitions below apply it to progressively broader sets of reported recall causes.

We compare three definitions that include different sets of Class I recall causes associated with failures of treatment safety or reliability. All three also include the same verified clinical safety or efficacy actions, including withdrawals. The narrow definition starts with sterility or biological-contamination problems, severe side-effects and dose or potency failures. These directly compromise two basic properties of the marketed medicine: its physical integrity and its delivery of the intended amount of active treatment. We therefore believe that they provide a transparent starting point for investigating interruptions to usable treatment.

The benchmark additionally includes serious identity, labeling and packaging errors, extending the criterion to failures that can cause the wrong medicine or treatment to reach the patient. The broader definition also includes stability, dissolution and other functional-quality failures, extending the comparison to defects affecting how the medicine maintains or delivers its intended action.

We use the intermediate definition as our exploratory benchmark, with the narrow and broad definitions providing sensitivity checks with other plausible definitions of recalls (see Table 4). Throughout the paper, we will often refer to this group of recalls as "major recalls".

A further concern in selecting recalls is that protection expiry can reduce sales independently of a recall, making it difficult to disentangle the effect of a recall from that of patent expiry on sales and leaving us uncertain about the recall's effective strength. This may happen in cases where the recall happens at the same time or right after the expiration of a patent, as illustrated in Figure 1.

[9]Their evidence concerns toys. It motivates examining possible spillovers, but does not establish their presence in pharmaceuticals.

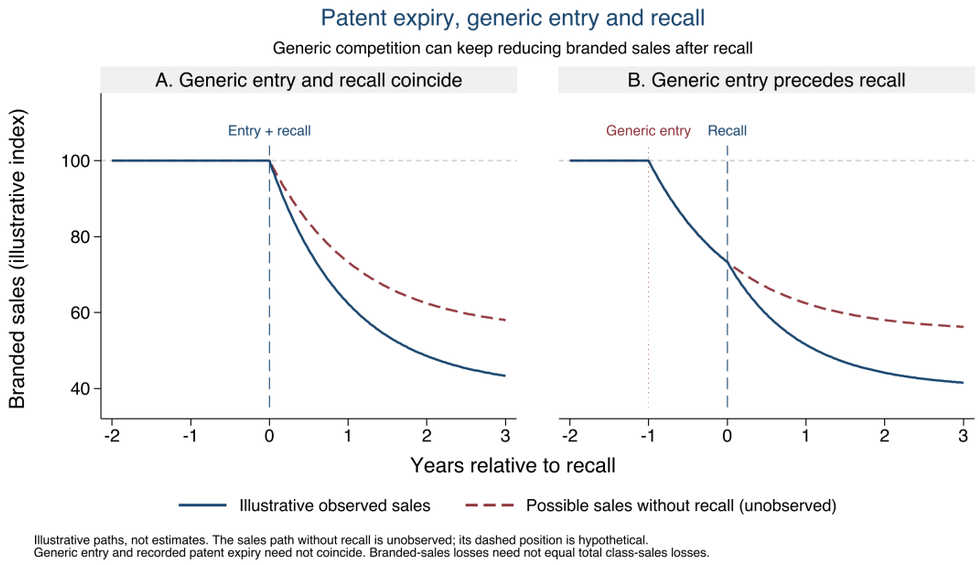


Figure 1: Generic entry and recall as overlapping sources of sales losses. Generic entry coincides with recall in the left panel and precedes it in the right panel. Solid lines illustrate branded sales with both events; dashed lines show hypothetical sales without recall. Continuing generic competition can therefore contribute to the decline observed after recall. The curves are illustrative, not estimates.

As a robustness check, we therefore restrict the brand comparison to products whose recorded expiry predates the entire observation window and repeat the preferred first stage excluding market-years near recorded expiries. The recall coefficient remains negative and very similar to the one of our main specification in Table 4 at both ATC levels. Namely, after excluding market-years near recorded expiries, the recall coefficients are -0.0153 (SE 0.0034) at ATC3 and -0.0161 (SE 0.0033) at ATC4, with effective F statistics of 20.07 and 23.70, respectively. These checks mitigate concerns about coincident recorded expiries, although incomplete historical protection information prevents ruling out expiry-related confounding entirely.

4.2 Data on innovation measure

We measure innovation using the total number of active trials at the ATC-3 (ATC-4) level, rather than traditional indicators such as cumulative R&D expenditures or the count of New Molecular Entities (NMEs). R&D expenditures are an input-based proxy for innovation and, in the pharmaceutical industry, they are known to be noisy and strongly shaped by accounting conventions and firms' financial and strategic choices. In particular, aggregate R&D at the market level tends to scale mechanically with the size and composition of the firms operating in that market,[10] so that using cumulative R&D spending as a proxy for innovation risks capturing market size itself rather than innovation effort. More importantly, R&D decisions are inherently forward-looking: they respond to current cash flows and to expectations about future market rewards. This intertemporal decision structure may induce persistence in unobservables and dynamic endogeneity, that is, shocks affecting investment at time $t-1$ can remain correlated with investment at time t through omitted expectations and financing conditions. To mitigate these concerns and in line with the literature, we measure innovation by the number of active trials at the ATC-3 (ATC-4) level, which captures actual experimental activity. Unlike expenditure-based measures, counts of active trials are less mechanically tied to contemporaneous sales and are less affected by accounting practices. In addition, our robustness checks explicitly address potential temporal dependence by including lagged innovation measures and testing for serial correlation in the error structure.

The choice not to rely on NMEs is also worth explaining. As in Acemoglu and Linn (2004), NMEs are innovative products containing active moieties that have not been pre-

[10]For instance, Stoneman (2010) and others have posited that smaller, newer market entrants may be more disposed toward R&D investment than their larger, established counterparts. Consequently, defining innovation merely in terms of a market's cumulative R&D spending could inadvertently reflect the market's size, as gauged by the aggregate sales of the participating firms.

viously approved by the FDA. Despite being comprehensive, NMEs do not fully capture firms' intentions to innovate because they are observed only at the end of a long and uncertain development and regulatory process. Pharmaceutical drug approval begins with a pre-clinical phase, followed by clinical trials.[11] Thus, NMEs are recorded only when products successfully complete trials and receive FDA approval (NDA classification). Because approval occurs several years after early research and clinical development, the timing of the effect of market size on NMEs is inherently lagged and potentially heterogeneous across therapeutic areas (Sertkaya et al., 2024), implying that NME-based specifications would require strong assumptions on the lag structure that are difficult to discipline in our ATC-3-by-year panel specification. Accordingly, our metric measures upstream innovation by counting active trials, capturing both successful and potentially unsuccessful projects (withdrawn or early-terminated), and providing a timelier measure of innovation effort. This approach also recognizes potentially unsuccessful trials as indicative of firms' commitment to innovation.

Our measure of innovation is, therefore, based on clinical trials across all Phases. This is a commonly adopted solution in the pharmaceutical literature (Cunningham et al., 2021; Gaessler and Wagner, 2022).
We collect trial records from ClinicalTrials.gov, supplementing the US extract with information retrieved through its API and linking records by their NCT study identifier. The initial sample retains interventional studies with at least one US site and a drug or biological intervention, across all recorded phases and sponsor types, without requiring successful completion or drug approval. We require a reported start year and exclude starts explicitly marked as estimated, studies not yet recruiting or withdrawn before enrollment, and records reporting zero actual enrollment. A new trial is counted in its reported start year and a completion date is not required.[12] Among other information on trials, we have also retained whether the trial was industry-sponsored or not.

We linked trials to ATC classes through the medicines listed as drug interventions in ClinicalTrials.gov. Drug names and aliases are matched exactly to products or single-ingredient medicines in our sales data, which supply the EPHMRA ATC3 and ATC4 codes. Ambiguous class assignments remain missing.[13] Each trial is counted once in each confirmed class. When listed medicines include comparators and no ATC information was available, we attribute the developing drug to the same ATC as the comparator one.

Over the period 2004–2016, the overall sample contains around 200000 clinical trials for U.S. drugs. The sample includes 43,959 newly initiated trials. Phase II accounts for the largest share of new trials (30.8%), followed by Phase I (24.0%), Phase III (16.6%), and Phase IV (10.4%). Early Phase I, combined phases, and trials classified as having no applicable phase account for the remaining 18.1%. Almost half of all new trials have an industry lead sponsor (48.1%), rising to 61.7% when trials with an industry collaborator are also included. Industry sponsorship is particularly common in Phase III, where firms lead 79.9% of newly initiated trials. The annual counts also show growth beyond 2013: industry-led active trials increase from 5,204 in 2013 to 6,030 in 2016, while new industry-led trials increase from 1,647 to 1,767.

Figures 2 and 3 describe interventional drug and biological trials registered on Clinical-

[11] In Phase 1, the company performs clinical trials on healthy volunteers to establish the drug's fundamental characteristics and safety profile in humans. According to DiMasi et al. (2003), *"typically, the drug remains in this stage for one to two years"*. Once the drug is dispensed to volunteers drawn from the targeted population, Phase 2 begins, and Phase 3 subsequently compares the new drug against a standard-of-care drug, thereby establishing efficacy and safety.

[12] Trial dates and eligibility fields use registry metadata dated 23 October 2026, retrieved through ClinicalTrials.gov API v2. These updated records retrospectively describe historical trials rather than reproducing the information available at the time. Historical records with an unspecified start-date type and studies with missing or inapplicable phase labels are retained, and missing phase information is not treated as preclinical research. Source and selection details are documented in `innovation_measure_management.do` and the saved extraction notes.

[13] Supplementary sources are NIH/NLM RxNorm, the product listings of the Irish medicines regulator (HPRA), and EPHMRA classification guidance, including its comparison with WHO codes. Full source URLs and individual classification decisions are recorded in the notes to `innovation_measure_management.do` and its accompanying mapping files. For the firm-level analysis, NCT identifiers link trials to IND lead-sponsor records; only unique sponsor-to-sales-company name matches are retained.

Trials.gov with at least one US study site. Both figures show all eligible trials and the subset with an industry lead sponsor. In Figure 2, Panel A counts each trial once in every year from its reported start through its reported completion, including both endpoint years. Panel B separately counts trials initiated in each year, distinguishing new research activity from studies already underway. Activity counts require both dates and include estimated completion dates; trials without a completion date remain included in the count of new starts. Figure 3 subtracts from annual activity the trials recorded as completed in that same year. Activity continues to increase with a peak around 2021–2022, which is not reported in the figures as the final year of the data at our disposal is 2015.

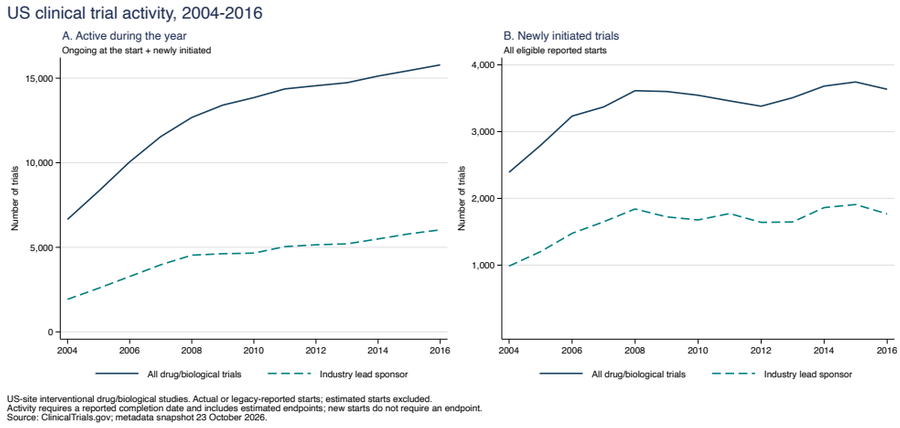


Figure 2: Active and newly initiated clinical trials with US sites, 2004–2025.

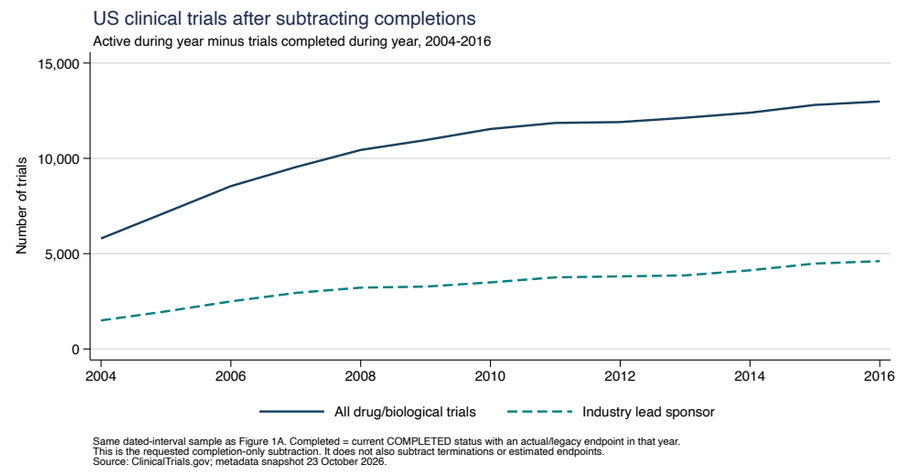


Figure 3: Active clinical trials with US sites, excluding trials completed in the same year, 2004–2025.

5 Methodology

This subsection explains the preferred methodology for exploring the causal effect of market size on innovation.

Our baseline empirical specification is Equation (6) that links innovation at the ATC-3-year level to market size through a Poisson model:

$$E[N_{mt}|\mu_m, \zeta_t, X_{mt}, M_{mt}] = \exp(\beta_1 \log M_{mt} + \beta_2 X_{mt} + \mu_m + \zeta_t), \quad m = 1, \ldots, N,\ t = 1, \ldots, T, \tag{7}$$

where N_{mt} is the number of active trials in ATC-3 class m and year t, M_{mt} is class-level market size measured by sales, X_{mt} is a vector of controls, μ_m are ATC-3 fixed effects, and ζ_t are year effects. In our application, X_{mt} includes variables capturing product age, competition, generic penetration, and scientific production.

Estimating (7) by treating $\log M_{mt}$ as exogenous may yield biased estimates for two reasons. First, market size is an equilibrium outcome and may therefore be correlated with

unobserved profitability shocks that also affect trial activity. Second, in a nonlinear count model, this endogeneity must be handled jointly with fixed effects. For this reason, we adopt the control-function IV approach of Lin and Wooldridge (2019), which is specifically designed for nonlinear models with fixed effects and allows one to address both endogeneity arising from time-invariant heterogeneity and endogeneity arising from time-varying idiosyncratic shocks.

Formally, let c_m denote time-invariant unobserved heterogeneity and let κ_{mt} denote an idiosyncratic market-year shock. The conditional mean can be written as

$$E[N_{mt} \mid M_{mt}, z_{mt}, c_m, \kappa_{mt}] = c_m \exp(x_{mt}\beta_1 + \kappa_{mt}), \tag{8}$$

where $x_{mt} = (M_{mt}, z_{mt})$ and z_{mt} contains the exogenous controls, including year effects. The key concern is that M_{mt} may be correlated with κ_{mt}. In the absence of such idiosyncratic endogeneity, a fixed-effects Poisson estimator would be consistent under the strict exogeneity condition

$$E[N_{mt} \mid x_{mt}, c_m] = c_m \exp(x_{mt}\beta_1).$$

The control-function approach allows us to test this restriction directly.

To implement the procedure, we first estimate the reduced form for the endogenous regressor. Let R_{mt} denote the excluded instruments, namely contemporaneous and lagged normalized major recalls. The reduced form for market size is

$$M_{mt} = z'_{mt}\Pi + R'_{mt}\delta + c_{m2} + u_{mt2}, \tag{9}$$

where u_{mt2} is the reduced-form error. We then obtain the within residuals

$$\ddot{u}_{mt2} = \ddot{M}_{mt} - \ddot{z}'_{mt}\hat{\Pi} - \ddot{R}'_{mt}\hat{\delta}.$$

In the second step, we estimate a fixed-effects Poisson model augmented with the first-stage residual:

$$E[N_{mt} \mid M_{mt}, z_{mt}, c_m, \hat{\ddot{u}}_{mt2}] = c_m \exp(x_{mt}\beta_1 + \hat{\ddot{u}}_{mt2}\rho). \tag{10}$$

The coefficient ρ captures idiosyncratic endogeneity. A robust Wald test of $H_0 : \rho = 0$ therefore provides a direct test of whether market size can be treated as conditionally exogenous. When $\rho \neq 0$, the inclusion of $\hat{\ddot{u}}_{mt2}$ delivers consistent estimation of β_1 under the assumptions in Lin and Wooldridge (2019). Intuitively, the control function isolates the component of market size that is correlated with the unobserved idiosyncratic shock, so that the remaining variation in M_{mt} can be interpreted causally. A more detailed derivation for our case, including the Mundlak representation that rationalizes this control-function correction in the presence of fixed effects, is reported in Appendix C.

A practical implication of the fixed-effects Poisson estimator is that the dependent variable must be nonzero for at least one time period within a panel unit. We therefore exclude ATC-3 classes that never record positive trial counts over the sample period. These account for about 10% of the ATC-3 categories in our data.

Recalls as an instrument for market size. To identify exogenous variation in market size, we instrument M_{mt} with normalized major recalls. Let R_{mt} denote the number of major recall events in class m during year t, and define

$$\tilde{R}_{mt} \equiv \frac{R_{mt}}{|\mathcal{J}_{mt}|} \cdot 100,$$

that is, recalls normalized by the number of products in the ATC-3. This normalization is important because broader markets mechanically have more opportunities for recalls; using recall intensity rather than raw recall counts removes this source of spurious correlation between recalls and market size.

The relevance of the instrument follows directly from the conceptual framework in Section 3. A major recall affects class-level sales through two channels. The first is a product-removal channel: the recalled drug exits, or becomes partially unavailable in, the set of marketed products, thereby reducing effective class-level variety and quality. The second is an informational channel: major recalls convey negative information about safety and compliance and may reduce prescribing, continuation, and adoption at the therapeutic-class

level. When within-class substitution is not perfect and/or the event carries reputational content, both channels reduce class-level revenues M_{mt}.

This mechanism also implies persistence in the first stage. A recall may depress sales contemporaneously, but generally its effect is experienced in subsequent periods (Freedman et al., 2012; Jovanovic, 2021; Liebman et al., 2023) if the recalled product is not immediately replaced or if the reputational shock dissipates only gradually. This is why lagged normalized recalls are included in the first stage. Section 5.1 elaborates on the instrument's relevance.

The fact that recalls originate from a supply-side disruption does not invalidate their use as an instrument in our setting. If substitution within the ATC-3 were perfectly elastic and recalls had no informational content, the sales loss from one recalled product would be immediately offset by reallocation to close substitutes, leaving class-level revenues largely unchanged. In that case, the instrument would be weak. Our identifying strategy instead relies on the empirically relevant case emphasized in Section 3: imperfect substitution and/or a reputational demand response imply that recalls reduce class-level revenues rather than simply reshuffling sales across incumbent products. Since our endogenous regressor is class-level sales, this is exactly the variation in market rewards that matters for innovation incentives.

5.1 Identification

As usual, the main threats to identification are the instrument's strength and the exclusion restriction.

The theoretical framework in Section 3 describes a possible channel through which recalls affect innovation: changes in product availability and perceived safety alter purchases and expected class revenues, thereby changing firms' incentives to invest. Perceived-quality responses are consistent with this channel when they influence innovation through these expected revenues. Our identification strategy additionally assumes that, conditional on the specified controls and fixed effects, the selected recall variation is uncorrelated with other determinants of innovation and that it affects trial activity through the revenue-related market rewards represented in the empirical model. Effects on approval probabilities, development requirements, or competitive opportunities beyond this revenue channel would threaten the exclusion restriction. The discussion and empirical checks below assess specific threats to this interpretation.

Namely, in what follows, we denote "S." as the concerns related to the Strength of the instrument and "ER." as the Exclusion Restriction: (S.i) It is important to distinguish between the recall of an off-patent generic product, for which many substitutes exist and for which entry barriers are low, versus a branded product. (S.ii) Lower current sales need not imply lower expected future revenues. As an example, generic recalls could increase a branded firm's expected revenues in the class if doctors and patients view generics as less substitutable due to quality risks. (ER.i) A recall may signal a lower likelihood of winning FDA approval, either because the recalled product shares characteristics with products in development or because the FDA increases its scrutiny of all products in that class. Or, a recall may signal less competition in the future. (ER.ii) Another possible concern is that clinical safety news could directly affect regulatory scrutiny and decisions.

A further type of criticism might concern the instrument's endogeneity (arguably similar to the comments on the exclusion restriction). The most representative of such concerns arise from the fact that it may be less attractive to innovate in markets where generic presence is particularly heavy. Since the majority of recalls arise from generic products, there is the (correlational) possibility that ATC markets where most generic recalls happen are systematically the less innovative ones. We will show in Section 5.2 that the proportion of generic recalls equals the total proportion of generics in the sample, consistent with recalls occurring at random. Namely, Figure 4 examines whether recalls occur in markets with different pre-existing research activity, after accounting for market characteristics and ATC and year effects. Such selection could link recalls to innovation through factors other than market sales, raising a concern for the exclusion restriction. The broad recall definitions show little evidence of remaining differences. For the recall-only component of our intermediate benchmark, excluding withdrawals, the adjusted distributions of preceding-year trial counts differ between recall and no-recall years at ATC3 when this comparison is considered alone ($p = 0.015$). However, the figure contains eight comparisons: four action definitions at two ATC levels. Accounting for these multiple tests raises the p-value to 0.120. The evidence of a difference is therefore sensitive to how these comparisons are assessed. Moreover, the

curves cross, so the result does not consistently indicate that recalls occur in more innovative markets. This comparison excludes the withdrawals included in our benchmark instrument. The curves also cross, rather than consistently indicating more research.

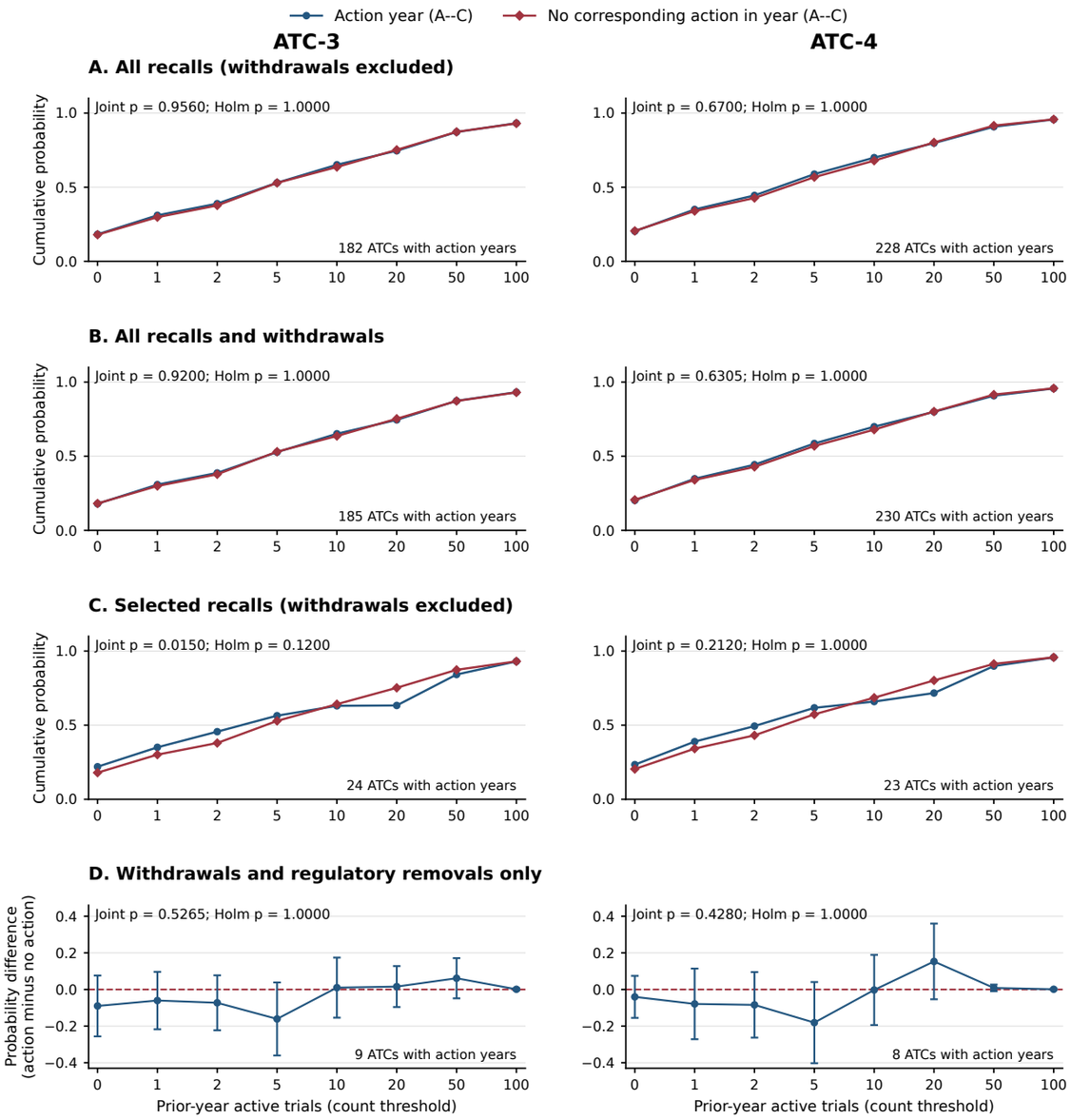


Figure 4: Recalls and pre-existing clinical-trial activity

Notes: Columns show ATC3 and ATC4 markets; rows distinguish four action definitions. In A–C, blue and red curves compare adjusted distributions in action and no-action years. At each threshold, the curve gives the probability of having no more than that many active linked drug trials in the preceding year; a higher curve therefore means fewer trials. D shows withdrawal-minus-no-withdrawal probability differences with 95% pointwise confidence intervals. Action status varies annually over 2006–2012: the same ATC can contribute to both groups. Adjustment includes ATC and year effects and preceding-year log product count, product and firm ages and their squares, generic share and concentration. Selected recalls retain Class I sterility, dose and label causes plus verified clinical recalls, excluding withdrawals. Withdrawals include verified noncommercial regulatory removals. Joint tests allow dependence within ATCs; Holm p-values adjust for eight comparisons. Trials cover all recorded phases; trial linkage is incomplete.

Table 1, moreover, examines whether firms start more clinical trials in the same therapeutic class after recalling a product. An increase could indicate an attempt to replace the recalled product through further research, a response that aggregate ATC trial counts might conceal. We compare years following at least one own-firm recall with other years, accounting for firm–ATC and year effects and the controls indicated in the table, with standard errors clustered by corporation. Withdrawals are excluded. Entries report percentage differences in expected new trial starts. For the preferred recall causes with pre-recall controls (row 2 in bold), the estimates are −65.9% at Firm–ATC3 and −57.3% at Firm–ATC4. The fact that both confidence intervals include zero excludes a surge in trials in light of a recall. The same exercise has been repeated for various definitions of recalls. Each uses (normalized) recalls in the previous year per 100 marketed products. The rows vary the Class I recall causes, counting

method, and controls with verified clinical safety or efficacy actions, including withdrawals, entering in every specification. Family counts group related recall notices into one issue, while event counts use distinct source identifiers. FE includes market and year effects, and AC adds age and competition controls. PRE uses these controls lagged by two periods. Our preferred benchmark is specification 2. Its Class I recalls concern severe biological contamination or safety concerns (e.g. failure to assure sterility).

The selected set also includes recalls and withdrawals prompted by concerns about the drug's safety in terms of severe side effects or effectiveness in clinical use.

Table 1: Recalls and subsequent trial starts within the same firm and therapeutic class

No.	Class I causes	Count unit	Controls	Firm–ATC3 Change (%) 95% CI	Firm–ATC4 Change (%) 95% CI
1	S, D, Q, L	Family	AC	−63.6* [-86.6, -0.9]	−43.7 [-79.9, 57.7]
2	**S, D, L**	**Family**	**PRE**	**−65.9 [-89.1, 6.2]**	**−57.3 [-83.1, 8.1]**
3	D, L	Family	FE	—	—
4	S, D, L	Family	AC	−63.6* [-86.6, -0.9]	−43.7 [-79.9, 57.7]
5	S, D, L	Family	FE	−65.6 [-89.3, 10.1]	−60.5* [-83.5, -5.2]
6	D, L	Family	PRE	—	—
7	D, L	Family	AC	—	—
8	S, L	Family	PRE	−57.7 [-82.3, 0.9]	−43.7* [-67.4, -2.7]
9	S, L	Family	FE	−56.9 [-81.6, 1.2]	−48.3** [-66.7, -19.7]
10	S, L	Family	AC	−54.7* [-76.6, -12.2]	−18.5 [-49.1, 30.7]
11	S, D	Family	AC	−52.2* [-73.4, -14.1]	−32.2 [-69.0, 48.2]
12	S, D	Family	FE	−52.6* [-76.4, -4.9]	−54.4* [-76.6, -11.2]
13	S, D	Family	PRE	−53.6* [-77.5, -4.4]	−50.1 [-75.9, 3.3]
14	L	Event	FE	—	—
15	L	Event	PRE	—	—
16	L	Event	AC	—	—
17	S, Q	Family	PRE	−37.9*† [-58.3, -7.5]	−31.6† [-55.3, 4.7]
18	D	Family	FE	—	—
19	D	Family	PRE	—	—
20	D	Family	AC	—	—

Notes: FE Poisson estimates report $100[\exp(\widehat{\beta})-1]$, the conditional percentage difference in expected annual newly started linked US DRUG trials when the same firm–ATC had at least one recall in the preceding year, versus none. Confidence intervals transform both log-coefficient endpoints; stars test $\beta = 0$ with asymptotic normal corporation-clustered sandwich inference: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Every finite cell has only one observed trial start in exposed years. † marks identification from one exposed corporation. Inference is therefore very fragile in that case. Dashes indicate no finite recall coefficient because exposed outcomes are all zero, not an estimated zero effect. Rows retain the 20 first-stage cause sets, count units and FE/PRE/AC timing, but use an any-recall indicator rather than a normalized count. Family/event choices yield the same indicator for otherwise identical source rules. S: sterility; D: dose/potency; Q: quality; L: labeling. Withdrawals, cessations, and suspensions are excluded; no clinical action qualifies in 2006–2012. All models include firm–ATC and year effects. FE adds no controls; PRE uses its own mean product age and its square, log product count, and generic share at $t-2$; AC uses current values, which may be affected by recall. Outcome years are 2007–2013, with positive own sales at $t-2$.

An additional argument on the latter issue – that recalls or withdrawals may themselves influence firms' innovation decisions, either by creating opportunities to fill a market gap or by inducing firms to postpone R&D investments in the affected therapeutic area – is that the timing of these effects is important. While a recall occurring at time t or $t+1$ is unlikely to affect innovation already underway at time t, a recall occurring at $t-1$ could, in principle, influence subsequent innovation decisions. Nevertheless, pharmaceutical innovation is a lengthy process, often requiring many years of development and regulatory

approval. Moreover, major recalls are relatively rare at the ATC level, and the substantial sunk costs of ongoing clinical trials make it unlikely that an isolated recall would prompt firms to significantly alter or abandon existing R&D projects. However, because investment requirements vary substantially across the different Phases of pharmaceutical development, the subsequent analyses also estimate the second-stage specification separately for each phase of the innovation process.

5.2 Strength of the instrument

Let us start from (S.i). Figure 5 provides descriptive evidence of the latter. In particular, it examines the alternatives available to 109 selected generics in the retail demand sample. In the left-hand panels, each dot compares a generic's number of rival presentations sharing its ingredient combination and form with the corresponding average for nongeneric products offered in the same market. Rivals are products of other corporations whose strengths may differ. Points above the diagonal indicate more recorded alternatives for the generic. Generics have, on average, 29.0 recorded ingredient/form rivals in ATC-3, compared with 11.3 for the nongeneric benchmark. The corresponding averages in ATC-4 are 28.5 and 11.2. Thus generics have more potential alternatives on average (although these counts do not establish clinical interchangeability or low entry barriers.)

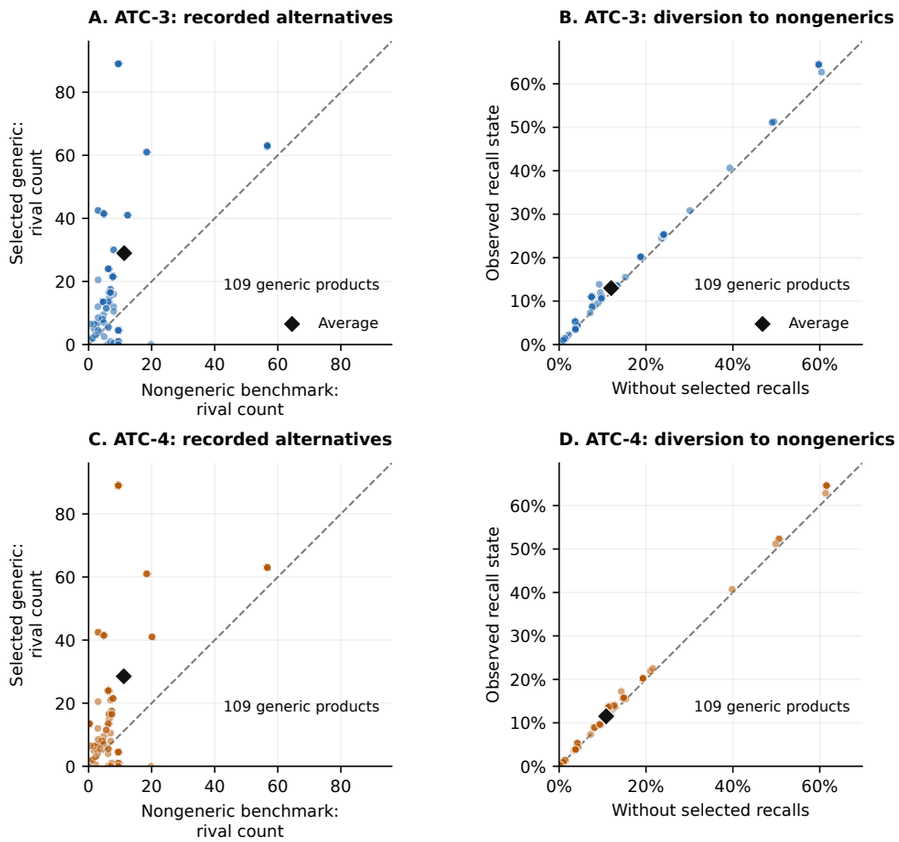


Figure 5: Recorded alternatives and predicted diversion for selected generics. Each dot represents a product, averaged over its active recall years. Left: ingredient/form rival counts relative to a nongeneric benchmark. Right: predicted diversion to nongenerics following a small generic-price increase, comparing no-recall and observed states at fixed prices and offerings. Diamonds show means; dashed lines indicate equality.

We complement these counts with a random-coefficients demand model in which buyers choose among available medicines or an outside option, and recalls can change product valuations and price responsiveness. Appendix D describes the model, estimation and recovery of elasticities. The right-hand panels ask where purchases lost following a small increase in a generic's price would go. Each dot compares the predicted fraction moving to nongenerics without major recalls with that in the observed recall state, holding prices and available products fixed. The point estimates imply a modest average change: approximately 12 out of 100 displaced purchases move to nongenerics without recalls, compared with 13 in the observed state in ATC-3; the corresponding figures are 10.8 and 11.5 in ATC-4. In other

words, an increase in price of a generics does not change substitution patterns to nongenerics.[14] More precisely, the fitted random logit model predicts that more price-displaced purchases remain within the therapeutic class in the observed recall state, predominantly moving to generics. The increase in diversion toward nongenerics is comparatively modest as shown in Table 2.

Table 2: Model-predicted destinations of purchases lost after a small source-price increase. Entries compare no selected recalls → the observed recall state.

Price increase → destination	ATC-3	ATC-4
Generic → other generics	69.8% → 76.7%	71.9% → 76.9%
Generic → nongenerics	12.0% → 13.0%	10.8% → 11.5%
Of which: brands labeled off-patent	5.3% → 5.9%	4.6% → 5.0%
Nongeneric → generics	70.2% → 77.8%	72.6% → 78.2%
Nongeneric → other nongenerics	11.5% → 12.8%	10.3% → 11.1%

Note: The indented row is included in total diversion to nongenerics. It covers nongeneric named brands recorded as "NO LONGER PROTECTED". This classification does not establish the date of patent expiry.

Appendix D, Figure D.1, complements the evidence in Table 2. The fitted model predicts a slightly smaller percentage quantity response to a price increase under observed recall exposure. This is consistent with Table 2: conditional on purchases being displaced, a larger fraction moves to other generics, while diversion towards nongenerics changes modestly. These findings describe different margins—i.e., how much demand leaves a product and where that demand goes. Consequently, many potential alternatives and substantial diversion towards generics do not necessarily imply high own-price sensitivity or complete replacement of recalled sales. Since these conditional predictions characterize substitution but do not rule out compensating revenue gains for competing products, we will separately assess whether recalls predict lower total class revenues through our class-level first-stage estimates.

The above, however, are descriptive statistics that do not control for relevant confounding variables and fixed effects. Our source data distinguish generic product type from brand status, including branded generics. Using classifications from presentations sold by the same corporation before the recall, roughly three-quarters of the selected products are generic. Notice, however, that the latter proportion is consistent with a random selection of recalls, since generics account for 78% of the products in the sample overall. Among exposures with baseline information, moreover, 46% of affected sales had at least five other recorded suppliers of the same ingredient/form. While this suggests that substitutes exist, it does not establish how much replacement actually occurred. We therefore examine whether recalls predict lower total class sales, including revenues earned by competing products, as shown in Figure 6.

[14]We report model-based point estimates without confidence intervals. Dubois, Griffith and O'Connell (2018, p. 420, footnote 17) construct Monte Carlo intervals by drawing jointly from the estimated asymptotic parameter distribution. Applying an analogous procedure here requires particular care because recall heterogeneity is imprecisely estimated and standard-deviation parameters are close to their zero boundary, making the usual normal approximation questionable. Validating such inference lies outside the scope of the present work.

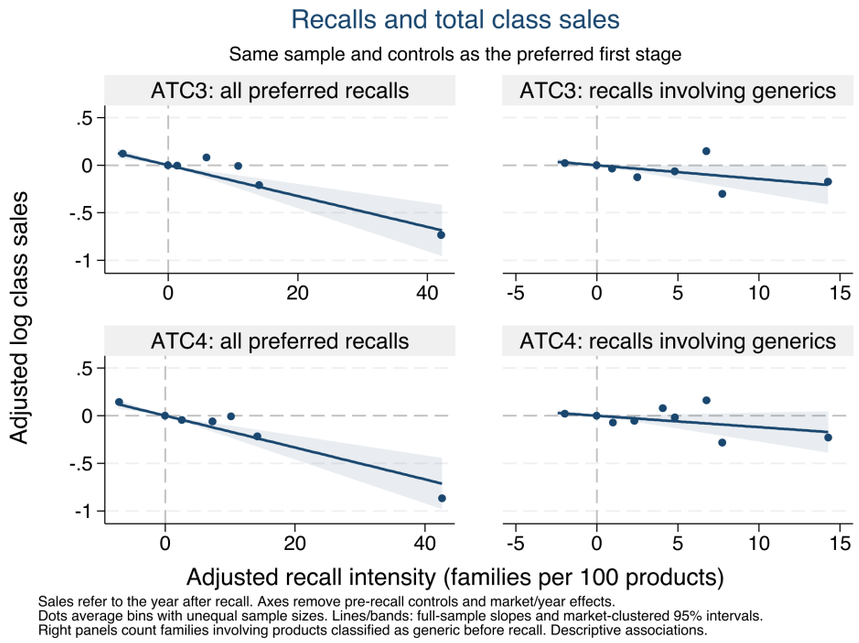


Figure 6: Recall intensity and total class sales in the following year. Both axes account for pre-recall controls, market effects, and year effects. Left panels use all preferred recalls; right panels consider recalls involving generics. Dots show binned averages, while lines show the estimated relationships with market-clustered 95% confidence bands.

Using the same sample and controls as the preferred first stage, the figure shows that greater recall intensity is associated with lower sales in the following year at both ATC3 and ATC4.

While Table 2 shows little evidence about generics-to-non-generics substitution, it is not informative about the persistence of recalls. To answer (S.ii), because innovation responds to expected future returns, we examine sales beyond the year used in the first stage. The preferred specification (in bold in Table 4) predicts lower average class sales in the second and third years after the recall, consistent with a persistent revenue component. A descriptive check of whether class sales recover quickly after selected recalls, as replacement by competing products could offset the initial losses, is presented in Figure 7. The latter relates the preferred recall intensity to class sales at ATC3 and ATC4. Year 0 begins in the recorded action quarter. Small markers show quarterly estimates, while diamonds summarize total sales over each four-quarter year.[15] Both curves use the same sample, pre-recall controls, and market and time fixed effects. Bars show pointwise 95% confidence intervals. Negative coefficients indicate that greater recall intensity is associated with lower sales.

[15]Annual estimates use the log of total four-quarter sales, rather than the average of quarterly log sales; their coefficients and intervals therefore need not equal averages of the quarterly results. In year 0, the first quarterly estimate is positive and imprecise; only the following three intervals lie entirely below zero.



Figure 7: Quarterly and annual class sales around selected recalls. Points report the association between the preferred recall intensity and log class sales at ATC3 and ATC4. Small markers represent individual quarters; diamonds summarize four-quarter sales totals. Year 0 begins in the recorded action quarter. Bars show pointwise 95% confidence intervals clustered by market and calendar quarter.

At both ATC levels, the three quarters following the action quarter show negative sales associations, as does the following four-quarter year. Later annual estimates remain negative but are less precise. These descriptive patterns suggest that losses can extend beyond the immediate market-level interruption. While the literature does not establish a universal time lag for pharmaceutical innovation decisions, Finkelstein (2004) documents an increase in vaccine clinical trial starts approximately 1 year after policies that increased expected returns were implemented.

To better understand if recalls affect future returns, we should revert to the first-stage controlling for the relevant covariates (among others) that we have identified so far as being potential sources of confounding, i.e., generic/branded status, market and year fixed effects, exposure to patent and exclusivity expiries known before the recall, and measures of generic penetration and market concentration (such as the HHI index). The latter is covered in Section 6.

5.3 Exclusion restriction

Exclusion restriction's threats are mainly described by (ER.i) and (ER.ii). Let's first unpack (ER.i), which guards against the possibility that a recall of a marketed product may trigger the probability of carrying on developing a medicine that is actually on trial or prevent the FDA from giving approval for the start of a new trial of a medicine similar to the one recalled. Since we do not have sufficient information to determine whether the recalls led the FDA to withhold approval to begin a new trial, we will examine only one implication of (ER.i). Namely, Figure 8 asks whether a recall of a marketed medicine is associated with more frequent early closure of ongoing trials studying medicines in the same ATC and similar characteristics. The concern is that recalls might interrupt development through channels other than their effect on market revenues. We identify the experimental medicine, excluding comparators and background treatments, and retain trials registered and started before the recall. We then record whether the trial closes within the following twelve months with status "terminated", distinguishing early closure from normal completion. Trials in affected classes are compared with eligible trials in other classes without a qualifying event on the same date. We account for therapeutic-class and comparison-date fixed effects, trial age, and pre-recall product counts and generic shares. The panels use ATC-3 and ATC-4 definitions and distinguish preferred events from all Class I recalls.

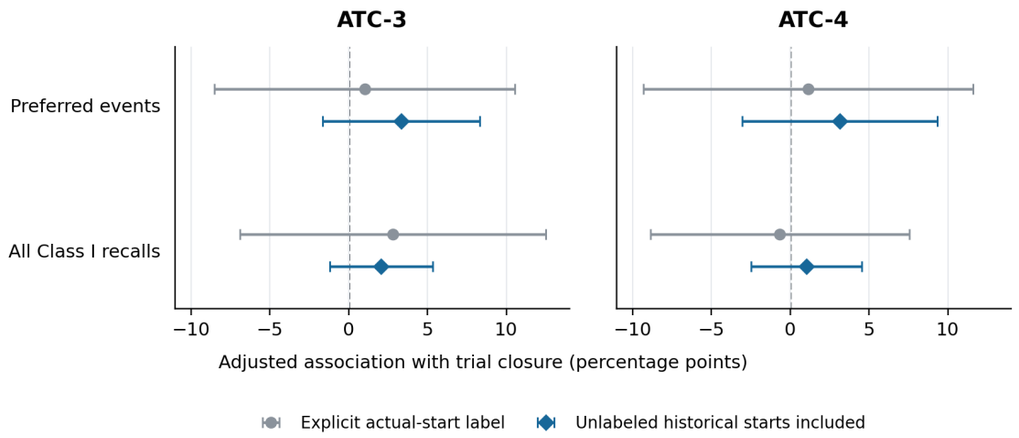


Figure 8: Marketed-drug recalls and trial closures recorded as terminated within twelve months in the same ATC. Points show adjusted percentage-point associations; bars show clustered 95% confidence intervals. Blue additionally retains unlabeled historical start dates; grey requires an explicit actual-start label.

Each point reports an adjusted percentage-point association, with a clustered 95% confidence interval. Positive values indicate more frequent closure following exposure. Blue retains historical start dates without an explicit "actual" label, while excluding estimated starts; grey requires that additional label. The broader sample generally improves precision. Although the point estimates are positive, every interval includes zero and meaningful increases, so the figure does not establish either an increase or a negligible association. Importantly, the data do not establish whether termination was mandated by the FDA or decided by the sponsor or investigators. Closures could reflect regulatory concerns, supply problems or commercial considerations, including changes in expected revenues. Nor does closing one trial necessarily mean abandoning the medicine's development.

(ER.ii) concerns the possibility that safety news affects regulatory scrutiny and decisions. A similar concern is expressed in Grennan and Town (2020) for medical devices. We are, namely, interested in the changes in innovation induced by pharmaceutical recalls (and withdrawals) that are not mediated by market size. The latter, indeed, does not invalidate the instrument. In particular, we are interested in unobservable (and observable) potential confounders that do link recalls directly to our innovation measure. For example, suppose a recall prompts the FDA to require an additional safety study for medicines under development. That requirement can discourage innovation even if class sales remain unchanged.

To look for observable signs of this concern, we examine whether trial activity for other medicines and firms changes around a recall, alongside revenues in the affected therapeutic class. Additional regulatory requirements could delay new studies, prolong existing ones or discourage development. We therefore distinguish new trial starts from studies documented as unfinished, rather than treating every change in the latter as abandonment [16]
More precisely, let N_{mt}^{other} denote new trial starts for other medicines and firms in therapeutic class m. We examine how this outcome changes around a qualifying recall in that class for medicines and corporations different from the medicine recalled. Importantly, the exercise aims to compare the effect of the instrument R_i across trials of similar products, i.e., on N_j for all $j \in m$ with $j \neq i$. Hence, since if there are recalls on i it is very likely that there are also for the other $j \in m$ (Collins et al., 2013; Krieger et al., 2022), if we had an effect guided solely by the mechanism described in ER.ii then we should expect to find nonzero effects, where instead – as we will see – we find no effects.

We follow ATC-3 classes for four quarters before and after their first qualifying event,

[16]To complement this comparison, we use AERS reports of suspected adverse reactions, dated when the FDA received each report, to examine safety reporting before and around recalls. In separate checks, we ask whether therapeutic classes that subsequently experience a recall already had more reporting than classes without a qualifying event, and whether reporting about other marketed medicines in the affected class changes around the recall. We also follow reports involving the recalled ingredient alongside class sales and trial activity. In both cases, we found no significant results. The analyses are available upon request.

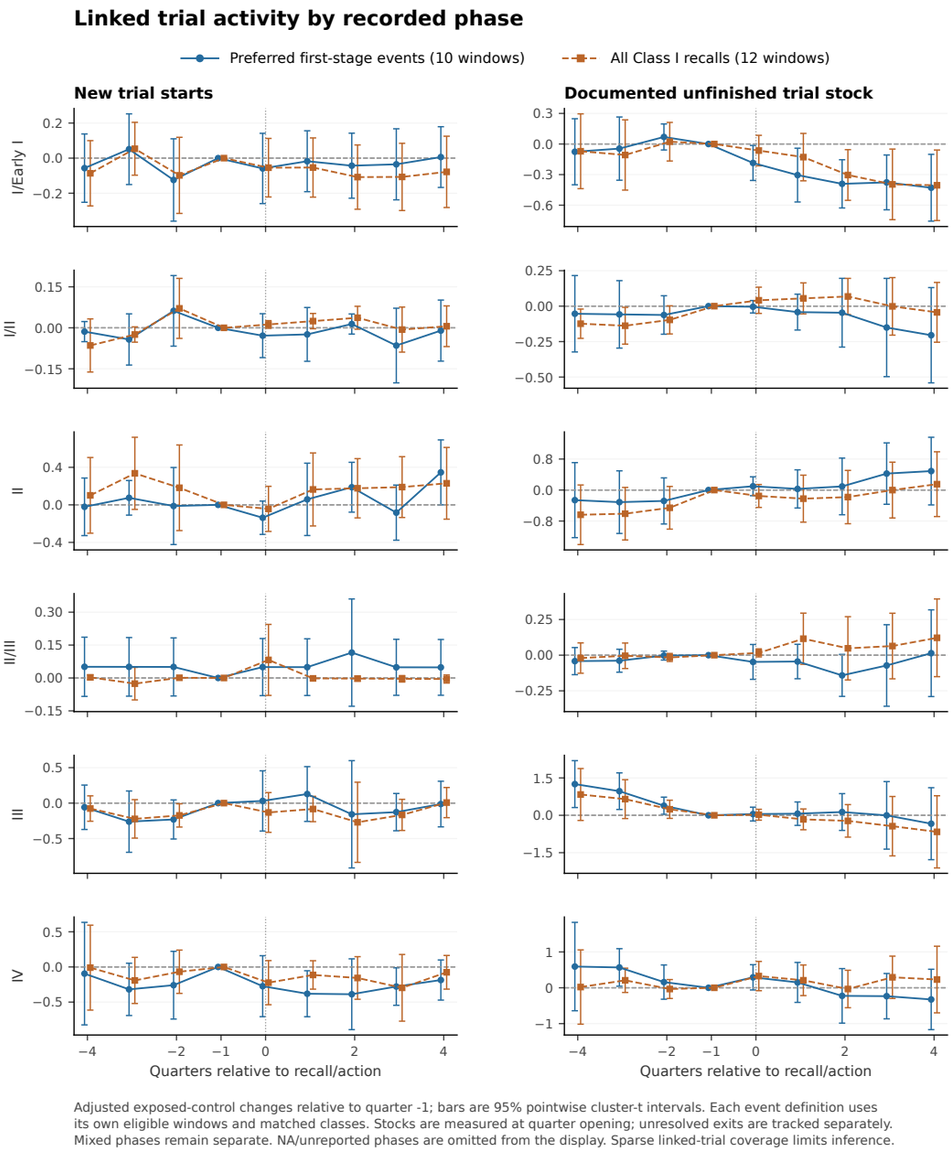


Figure 9: Trial activity by recorded phase around preferred first-stage events (blue; ten eligible windows) and all Class I recall causes (orange; twelve). Left: new trial starts. Right: documented unfinished studies at quarter opening. Points show adjusted changes in affected classes relative to matched comparison classes and quarter −1; bars show 95% pointwise clustered confidence intervals. NA/unreported phases are omitted from the display.

separately using our preferred definition and all Class I recall causes. To focus on effects reaching other medicines and other firms, we exclude trials with verified links to the recalled ingredient or observed affiliations with the recalling firm. Affected classes are compared with matched classes in the same therapeutic group, accounting for persistent class differences, common calendar changes and pre-event characteristics.

Each estimate compares the affected class's change from the quarter before recall with the corresponding change in its comparison classes. A negative estimate therefore indicates a larger fall, or a smaller increase, in the affected class. Figure 9 shows these comparisons by recorded trial phase. Table 3 averages the estimates over quarters +1 to +4, excluding the quarter containing the action. Revenue is indexed to 100 in the earlier matching period, so its coefficients are index-point differences rather than percentage changes.[17]

[17] Companion figures showing ingredient-linked adverse-event reporting, class revenues and monthly reporting patterns are available upon request.

Table 3: Average adjusted differences during quarters +1 to +4 after the action.

Outcome	Preferred events	All Class I recalls
New starts per class-quarter	−0.25 [−0.83, 0.32]	−0.14 [−0.54, 0.25]
Revenue, baseline-index points	−0.54 [−21.30, 20.23]	19.96 [3.08, 36.84]
Documented unfinished studies	−0.11 [−1.68, 1.46]	−0.28 [−1.52, 0.96]

Notes: Entries average adjusted exposed–comparison differences relative to quarter −1. Brackets report 95% clustered confidence intervals for each average, without multiple-comparison adjustment. The samples contain ten preferred and twelve all-Class-I event windows. Unresolved completion dates are tracked separately rather than classified as completed studies.

Across both event definitions, we find no statistically detectable aggregate change in new trial starts after recall. We therefore find no clear evidence of a widespread decrease (nor increase) in trial entry among other medicines and firms within the observed year in our preferred specification of recalls. Documented phase-I unfinished stocks, however, decline under both definitions, but the underlying records include completed studies, some ending before recall. This cannot be interpreted as evidence of regulatory discontinuation or abandonment.

6 Results

The present section describes the results obtained by applying the CF-Poisson two-stage approach of Lin and Wooldridge (2019). The section is divided into two subsections corresponding to the first and second -stage of the procedure respectively.

6.1 First-stage results

Given the above considerations, Table 4 reports twenty selected first-stage specifications with recall coefficients for log annual sales at the ATC3 and ATC4 levels, controlling for covariates, including generic and protection status, which we found to be important shifters of recalls.

Namely, Table 4 relates log annual class sales to selected recall exposure in the preceding year, normalized per 100 products marketed in that year. Rows differ in the Class I causes included, whether related notices are grouped into issue families or counted by source event, and the controls used. Verified clinical safety/efficacy actions, including withdrawals, enter every specification. All regressions include market and year fixed effects, with standard errors clustered by market. FE adds no further controls. AC includes current mean product age, a portfolio-based firm-age proxy, their squares, sales concentration (HHI) and the share of products with generic-type presentations. PRE uses the same variables measured at $t-2$, before recall exposure at $t-1$. FINAL retains PRE and adds firm count, log product count, publication growth $\Delta \log(1+P)$ and its square, also measured at $t-2$, where P counts medicine-indexed PubMed articles. These additions account for supplier presence, product variety and scientific activity. FINAL requires complete publication mapping and retrieval, so its comparison with the other rows also reflects differences in sample coverage.
Our preferred specification is "FINAL" highlighted in bold.

Table 4: First-stage estimates: normalized recalls and market sales

No.	Class I causes	Count unit	Controls	$\widehat{\beta}_{t-1}$	F_{eff}	Products	Families
Panel A: ATC3							
1	Sterility, Dose, Quality, Label	Family	AC	-0.0137^{***}	19.10	78	37
2	Sterility, Dose, Label	Family	PRE	-0.0162^{***}	24.84	75	34
3	Dose, Label	Family	FE	-0.0143^{***}	29.10	63	26
4	Sterility, Dose, Label	Family	AC	-0.0148^{***}	30.37	75	34
5	Sterility, Dose, Label	Family	FE	-0.0141^{***}	30.34	75	34
6	Dose, Label	Family	PRE	-0.0166^{***}	24.55	63	26
7	Dose, Label	Family	AC	-0.0150^{***}	29.38	63	26
8	Sterility, Label	Family	PRE	-0.0166^{***}	26.67	62	24
9	Sterility, Label	Family	FE	-0.0145^{***}	32.16	62	24
10	Sterility, Label	Family	AC	-0.0150^{***}	31.69	62	24
11	Sterility, Dose	Family	AC	-0.0150^{***}	33.28	54	26
12	Sterility, Dose	Family	FE	-0.0143^{***}	31.79	54	26
13	Sterility, Dose	Family	PRE	-0.0164^{***}	25.82	54	26
14	Label	Event	FE	-0.0147^{***}	30.70	50	16
15	Label	Event	PRE	-0.0171^{***}	26.22	50	16
16	Label	Event	AC	-0.0152^{***}	30.60	50	16
17	Sterility, Quality	Family	PRE	-0.0153^{***}	17.86	44	19
18	Dose	Family	FE	-0.0145^{***}	30.49	42	18
19	Dose	Family	PRE	-0.0168^{***}	25.45	42	18
20	Dose	Family	AC	-0.0152^{***}	32.35	42	18
21	**Sterility, Dose, Label**	**Family**	**FINAL**	$\mathbf{-0.0208^{***}}$	**50.49**	**18**	**9**
Panel B: ATC4							
No.	Class I causes	Count unit	Controls	$\widehat{\beta}_{t-1}$	F_{eff}	Products	Families
1	Sterility, Dose, Quality, Label	Family	AC	-0.0136^{***}	11.27	77	36
2	Sterility, Dose, Label	Family	PRE	-0.0167^{***}	27.27	74	33
3	Dose, Label	Family	FE	-0.0161^{***}	25.25	62	25
4	Sterility, Dose, Label	Family	AC	-0.0152^{***}	22.73	74	33
5	Sterility, Dose, Label	Family	FE	-0.0153^{***}	20.20	74	33
6	Dose, Label	Family	PRE	-0.0173^{***}	28.77	62	25
7	Dose, Label	Family	AC	-0.0147^{***}	15.55	62	25
8	Sterility, Label	Family	PRE	-0.0176^{***}	41.95	61	23
9	Sterility, Label	Family	FE	-0.0159^{***}	25.54	61	23
10	Sterility, Label	Family	AC	-0.0156^{***}	24.84	61	23
11	Sterility, Dose	Family	AC	-0.0158^{***}	30.60	53	25
12	Sterility, Dose	Family	FE	-0.0151^{***}	30.48	53	25
13	Sterility, Dose	Family	PRE	-0.0176^{***}	44.18	53	25
14	Label	Event	FE	-0.0168^{***}	35.30	49	15
15	Label	Event	PRE	-0.0183^{***}	49.86	49	15
16	Label	Event	AC	-0.0151^{***}	17.06	49	15
17	Sterility, Quality	Family	PRE	-0.0163^{***}	17.88	43	18
18	Dose	Family	FE	-0.0171^{***}	46.68	41	17
19	Dose	Family	PRE	-0.0183^{***}	54.30	41	17
20	Dose	Family	AC	-0.0154^{***}	21.05	41	17
21	**Sterility, Dose, Label**	**Family**	**FINAL**	$\mathbf{-0.0199^{***}}$	**41.63**	**22**	**13**

Notes: The dependent variable is log annual market sales. Every row uses one excluded instrument: recalls at $t-1$, normalized as 100 times the selected count unit divided by marketed products before lagging. Every set also includes verified clinical safety/efficacy actions, including withdrawals. Class I causes: sterility includes biological contamination; dose includes strength/potency; quality includes performance/stability; label includes identity/packaging. Family denotes an issue family; Event denotes an FDA/source-event identifier. Products and families are distinct affected counts.
All models include market and year fixed effects, with standard errors clustered by market. FE adds no other controls; AC adds mean product age, the portfolio-based firm-age proxy, their squares, HHI and generic share; PRE uses their $t-2$ values. FINAL retains PRE and adds firm count, log product count, publication growth $\Delta \log(1+P)$ and its square, also measured at $t-2$. FINAL requires complete publication mapping and retrieval.
Outcome years are 2007–2013. Rows 1–20 use 2,044 observations and 292 markets at ATC3, and 3,099 observations and 443 markets at ATC4. FINAL uses 602 observations and 86 markets at ATC3, and 1,148 observations and 164 markets at ATC4. Differences in F_{eff} may also reflect changes in sample composition. With one instrument, F_{eff} equals the squared clustered t-statistic. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$: nominal post-search significance. Acemoglu–Linn (2004) and Finkelstein (2004) motivate the mechanisms. Further definitions and source criteria are documented in the replication materials.

The analysis reveals that the effect of recalls at the ATC-3 level is substantial and significant, in line with Higgins et al. (2021) who document changes in firms' R&D portfolios following FDA safety advisories. This finding is robust to a wide range of covariate specifications and fixed effects, as well as to alternative definitions of recalled medicines. The F-statistic remains above 10 for all the specifications. We partly ascribe the robustness of these findings to the chosen levels of aggregation, i.e., ATC-3 and ATC-4. Firms characterized by high-quality management and a readiness for risk can promptly address severe recalls. Nevertheless, such recalls frequently surprise ATC-3 markets. Consequently, competitors fail to anticipate severe recalls impacting firms within the same ATC, an occurrence that is only recognizable at the market level.

6.2 Second-stage results

Table 5 reports the relationship between market size and clinical-trial activity over 2007–2015. Years 2004-2006 were excluded because of too many zeros in the dependent variable. The specifications progressively account for differences across therapeutic classes and the endogeneity of sales. Columns (P) and (NB) provide pooled Poisson and negative-binomial benchmarks. Namely, Column (P) features a simple Poisson model that uses exogenous market size to assess the robustness of market size's positive effect, without fixed effects or controls for endogeneity. Due to observed overdispersion in the data, Column (NB) introduces a naive Negative Binomial model to accommodate this variance. Column (FE-P) includes market fixed effects, allowing persistent class characteristics to be correlated with sales and the other regressors. Column (CF-IV), the preferred specification, additionally introduces the control-function correction described in the methodology section. All four specifications include year effects and the FINAL controls.
Notice that our IV estimates are primarily informed by recall-induced contractions in class-level revenues. Interpreting the estimated elasticity as the response to market expansions requires the maintained assumption that trial activity responds with the same elasticity to increases and decreases in $M_{mt} = \log(\text{Sales}_{mt})$. The constant-elasticity specification imposes this symmetry [18].

Table 5: Market sales and linked trial stock: ATC-3

	(P)	(NB)	(FE-P)	(CF-IV)	A–B log(1+N)	(NR)
Log sales	0.4868**	0.1604*	0.0808*	0.7886**	1.3828*	0.8060**
	(0.1566)	(0.0656)	(0.0352)	(0.2800)	(0.5444)	(0.2960)
First-stage residual	—	—	—	−0.7111**	—	−0.7282*
				(0.2632)		(0.3025)
$\log(1+N_{m,t-1})$	—	—	—	—	0.6344	—
					(0.9301)	
FINAL	Yes	Yes	Yes	Yes	Yes	No
Year effects	Yes	Yes	Yes	Yes	Yes	Yes
Class effects	No	No	Yes	Yes	Differenced	Yes
Observations	639	639	630	630	639	630
Classes	71	71	70	70	71	70
First-stage observations	—	—	—	639	—	639
First-stage clustered F	—	—	—	58.419	—	129.561
Recall-exposed Poisson classes	—	—	—	8	—	8
Residual-null test p	—	—	—	0.041	—	0.010
AR(1) p	—	—	—	—	0.000	—
AR(2) p	—	—	—	—	0.475	—
Hansen p	—	—	—	—	—	—
GMM instruments	—	—	—	—	21	—

Notes: 2007–2015. Stock counts linked start–completion intervals. P/NB are pooled Poisson/negative-binomial models; FE-P and CF-IV include class effects; NR omits FINAL. FINAL denotes age, competition, generic-share and publication controls at $t-2$. CF-IV/NR instrument sales using selected issue families and clinical actions per 100 marketed products at $t-1$. Standard errors are ATC-clustered; CF-IV/NR account for both estimation stages. A–B uses linear difference GMM with collapsed lag-2 outcome instruments; its log-sales coefficient is not a Poisson elasticity. Missing Hansen tests indicate exact identification. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$, unadjusted for specification search.

In Table 5, the coefficients of interest are Log sales and residuals: Log sales reflect market size, while residuals quantify the endogeneity of market size. A significant coefficient for residuals indicates a correlation between the error term. Endogeneity is tested using a Wald test on the coefficient of residuals, denoted "First-stage residual" in the Table and ρ in Section 5. As explained above, if ρ is significantly different from zero, endogeneity is confirmed, which is the case in our main specification, Column (CF-IV).
In particular, in Column (CF-IV), the estimated sales elasticity is 0.7886. Under the identifying assumptions discussed in Section 5.1, a 10% increase in market sales therefore implies approximately 7.8% higher expected trial stock. This estimate suggests that clinical research responds substantially, although less than proportionately, to changes in market rewards. As noted in Dubois et al. (2015), *"That the relationship of market size to innovation should be increasing seems obvious [...] However, it is not so obvious why entry should be less than proportionate. Many factors might play a role. One is decreasing margins as competition intensifies. Another is that as more firms are attracted into a market, there is a greater*

[18] Our design does not separately establish the response to positive market-size shocks.

chance that some of them duplicate each other's research efforts. [19]
Because N_{mt} counts linked studies whose reported start–completion intervals overlap year t, changes in the stock can arise through both new entry and the duration of existing studies. The estimate consequently describes the response of this measure of research activity, without separately identifying the contribution of each margin. Accounting only for newly entered studies, however, leaves the relationship quite unchanged [20].

The comparison across columns shows why accounting for endogeneity matters in this application. The sales coefficient is 0.4868 in the pooled Poisson model, 0.1604 in the negative-binomial model and 0.0808 in the fixed-effects Poisson model. Introducing the control function produces a substantially larger estimate. Moreover, the residual-null test rejects conditional exogeneity at the 5% level, providing evidence in favor of the correction under the maintained model assumptions. These comparisons indicate sensitivity to the treatment of endogeneity. Some empirical evidence is also replicated in the included covariates. For example, the coefficients on the average age of firms and its square are in line with past observations (Balasubramanian and Lee, 2008; Huergo and Jaumandreu, 2004). This effect evidences how the oldest firms tend to introduce less innovation than entrants in their early years. However, firms above intermediate ages appear almost as active in process innovations as entering firms, and even more in product innovations (Huergo and Jaumandreu, 2004).

Column (NR) examines the sensitivity of the preferred estimate to the observed controls by removing FINAL from both stages on the same eligible observations. The sales coefficient, 0.8060, remains close to the preferred estimate. Thus, within this sample, adding the age, competition, generic-share and scientific-activity controls has a modest effect on the estimated elasticity.

Column (A–B) considers persistence in trial activity by including $\log(1+N_{m,t-1})$ in a linear difference-GMM specification. The latter is an important specification since including lagged trial activity addresses the concern that persistent changes in scientific opportunities may influence current research. This follows the reasoning of Acemoglu and Linn (2004), who examine lagged entry and find statistically insignificant coefficients for nongeneric drug approvals and new molecular entities. The idea is that future demand for medicines is difficult to predict, as new therapeutic needs may emerge independently of the drugs under development in the preceding period. Consistently, in our A–B specification, the coefficient on $\log(1+N_{m,t-1})$ is also not significant.
In the (A-B) specification, the sales coefficient remains positive, whereas the lagged-outcome coefficient is imprecisely estimated. Here, the sales coefficient measures changes in $\log(1+N_{mt})$, so its magnitude requires a different interpretation from the Poisson elasticity. As suggested in Roodman (2009), *"in the context of an Arellano-Bond GMM regression, which is run on first differences, AR(1) is to be expected, and therefore the Arellano-Bond AR(1) test result is usually ignored in that context"*. Moreover, this output presents no significant evidence of serial correlation in the first-differenced errors at orders 2, 3, and 4 (the latter are not reported for the sake of space).
Overall, our results are consistent with the positive relationship between market rewards and pharmaceutical innovation documented in earlier research. Dubois et al. (2015) estimate an elasticity of approximately 0.23 for new chemical entities reaching the market and discuss several earlier estimates near 0.5. The higher effect observed in our analysis, compared to studies that use New Molecular Entities (NMEs) or New Chemical Entities (NCEs) as dependent variables, can be explained by the substantial costs associated with developing new pharmaceutical entities. Our approach assumes that active trials, by virtue of being ongoing and not yet concluded, include those most likely to contribute meaningfully to pharmaceutical innovation. We believe that our subset provides a more accurate reflection of the dynamic relationship between market size and the generation of valuable new treatments, thereby partially mitigating the issue of overestimation found in prior research. Drug development is, in fact, quite expensive, the cost ranging between $800 million and $2.5 billion (see, for instance, the FDA program MedWatch). Undertaking clinical trials is, instead, sensibly cheaper, amounting to an average of $20 million up to $40 million (Martin et al. (2017), John

[19] Our FINAL controls account for pre-existing differences in concentration, supplier presence, product variety and generic penetration. The demand analysis in Appendix D additionally examines substitution among marketed medicines. Identifying how these competitive relationships affect research incentives would require information on development costs, expected margins and competing research projects. In particular, we do not directly measure the duplication of research efforts across firms.

[20] Estimates are available upon request.

Hopkins Bloomberg School of Public Health, 2018). Thus, it is reasonable to suppose that, ceteris paribus, a 10% increase in market size stimulates more trials than NMEs or NCEs (on average). However, exceptions are still present (Acemoglu and Linn, 2004; Duggan and Scott Morton, 2010).
In an attempt to rationalize the difference between our estimate and the average one in the literature, we conducted an additional analysis utilizing solely industry-sponsored trials that culminated in a marketed innovation within the specified time frame, revealing a statistically significant coefficient of 0.2713 [21]. This examination revealed a possible overestimation bias concerning the elasticity of market size to innovation as presented in existing literature, possibly driven by the fact that existing estimates capture both the extensive and intensive margins of innovation. Specifically, market size not only influences the probability of success among initiated trials but also plays a critical role in firms' decisions to enter the innovation process in the first place. By focusing solely on marketed products, the analysis conditions on successful outcomes, thereby excluding failed or discontinued trials. As a result, part of the variation that reflects market size's role in encouraging the initiation of new trials is no longer captured. This restriction reduces the observed responsiveness of innovation to market size, leading to a smaller coefficient. Moreover, conditioning on successful trials may introduce a form of selection bias, as the sample no longer represents the full distribution of innovation attempts.

Clinical development stages differ in their financial commitments, remaining development costs and time to potential commercialization. Changes in expected market rewards may therefore affect the initiation of exploratory research and the continuation of later development differently (Alsan et al., 2024; Blume-Kohout and Sood, 2013; Gaessler and Wagner, 2022). In particular, exploratory Phase I studies and later registration trials involve different capital commitments, financial risks and remaining times to potential commercialization. Changes in expected revenues may therefore affect firms' willingness to undertake early research and pursue later development differently. This motivates separating the analysis by trial phase and focusing on studies with documented industry participation.

We repeat the preferred CF-IV specification separately for Phases I, II, III and IV among studies with an industry sponsor or collaborator. Each linked study counts once per therapeutic class in every year from its reported start through completion, including both endpoints, using its recorded phase. Combined phases, Early Phase I and unreported phases are excluded. We supplement missing sponsorship classifications with verified organizational information. If industry participation remains unresolved for any study contributing to a phase–class–year, that count is treated as missing.[22] All columns retain the preferred recall instrument, FINAL controls and market and year effects. Within each ATC level, the four models use the same full sales sample for the first stage, while the Poisson sample depends on the availability and within-class support of each phase outcome.

Table 6: Market sales and industry-linked trial stock by phase: ATC-3

	Phase I	Phase II	Phase III	Phase IV
Log sales	0.6364	−0.1029	1.6227**	2.0885**
	(0.7388)	(0.6523)	(0.6456)	(0.8115)
First-stage residual	−0.6387	0.2458	−1.5164**	−1.7573**
	(0.7387)	(0.6651)	(0.6708)	(0.7996)

Notes: 2007–2015. Preferred CF-IV with FINAL controls at $t-2$, recalls at $t-1$, and market and year effects. The common first stage uses 639 observations and 71 classes; clustered $F = 58.42$. Parentheses report ATC-clustered standard errors accounting for both estimation stages. Poisson observations/classes for Phases I–IV: 376/42, 441/49, 404/46 and 411/46. Stocks count reported start–completion intervals among studies with documented industry participation. Unknown sponsorship makes its phase/class-year missing. All-zero and uninformative singleton panels remain in the first stage but are omitted by FE Poisson. Phase IV is postmarketing. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; nominal tests without adjustment for multiple comparisons.

[21] Analyses available upon request.

[22] Industry participation uses the retrospectively recorded sponsor and collaborator information. We do not observe annual changes in sponsorship.

Table 7: Market sales and industry-linked trial stock by phase: ATC-4

	Phase I	Phase II	Phase III	Phase IV
Log sales	1.3598	−0.0587	2.2288**	2.2264*
	(0.9308)	(0.7420)	(0.8711)	(0..88101)
First-stage residual	−1.3349	0.1918	−2.1306*	−1.8631*
	(0.9258)	(0.7550)	(0.9335)	(0.9144)

Notes: Specification, outcome construction and inference follow Table 6, using ATC-4 markets. The common first stage uses 1,098 observations and 122 classes; clustered $F = 86.87$. Poisson observations/classes for Phases I–IV: 619/69, 720/80, 683/77 and 678/76.

Tables 6 and 7 show positive Phase I coefficients at both ATC levels and Phase II coefficients are not statistically significant. This is somewhat expected since, in practice, shifts in expected revenue are much more likely to affect late-stage clinical advancements (Phases III and IV) than on exploratory Phase I trials (Finkelstein, 2004; Gaessler and Wagner, 2022). For Phase III, the estimated elasticities are 1.62 at ATC3 and 2.23 at ATC4, both significant at the 5% level. These positive estimates are consistent with market rewards influencing later-stage development. Phase IV, concerning postmarketing research, also shows positive and significant estimates.

The phase-specific results clarify the economic activities underlying the relationship between market rewards and clinical research. The positive Phase III estimates show that this relationship is present in advanced clinical testing, even when postmarketing studies are excluded. The Phase IV estimates extend the evidence to research on medicines already on the market, a margin that other measures of innovation, such as counts of new molecular entities, do not directly capture. Differences between trial-based and approval-based elasticities may therefore reflect the different research activities covered, as well as the distinction between development effort and successful commercialization. This gives the trial-based measure a broader economic interpretation than the arrival of new medicines alone. The same qualitative pattern appears under both ATC-3 and ATC-4 market definitions. .

6.3 Robustness checks

The robustness analysis examines how the estimated relationship depends on the revenue variation used for identification and on the stage at which innovation is measured. Recalls can redistribute sales among medicines without substantially changing total revenues within a therapeutic class, making the choice of recall events important for instrument relevance. Trial activity and new-drug approvals also capture different stages of investment: approvals reflect development decisions made in earlier years and may respond to revenue changes over a different horizon. Comparing the estimates therefore requires attention to the strength of the first stage, the timing of the outcome and the research activities it covers.

Tables 8 and 9 examine the sensitivity of the results to the estimator, recall definition, instrument timing and innovation measure. All specifications retain the FINAL controls at $t-2$, year effects and controls for persistent differences across therapeutic classes. The first column estimates a dynamic model for $\log(1+N_{mt})$ using difference GMM. The next two columns broaden the recall definition and add current recalls to the preferred lagged instrument, respectively.

The last two columns consider alternative outcomes. The NME measure counts annual U.S. approvals of new molecular entities originally approved through New Drug Applications. Each approval counts once in every confirmed therapeutic class in its approval year.[23] The completed/terminated outcome restricts the original trial stock to studies recorded under either status on 23 October 2026, retaining their reported start–completion intervals. Completed studies ended normally; terminated studies stopped early.[24] These classifications are applied retrospectively and may reflect events after the sample period. Neither status establishes that a medicine received commercial approval. Both alternative outcomes retain

[23]The source is the FDA's compilation of NME and new biologic approvals. Of the 285 eligible approval records for 2004–2015, 283 are linked to the fixed 2015 EPHMRA ATC classifications used in the sales data. Products originally approved through Biologics License Applications are excluded. Twelve applications later converted administratively to biologics licenses are retained according to their original approval type.

[24]Definitions follow the ClinicalTrials.gov glossary.

the preferred sales first stage, although their second-stage samples depend on the availability and variation of each outcome.

We also made a robustness check with INDs. An Investigational New Drug (IND) application requests FDA authorization to investigate a drug or biological product in humans.[25] The IND dataset is part of the PHID data maintained at IMT and is only available as links at ATC-3. The two NCT-start columns in Table 8 compare current and one-year-lagged sales on identical market–year keys, each covering 2045 study starts. We have excluded to preserve coverage of research activity across therapeutic markets[26].

Table 8: Market size and innovation: robustness checks, ATC-3.

Outcome	(Two-step GMM) $\log(1+N)$	(Second-stage all recalls) N	(CF-IV) recalls $t, t-1$ N	(NMEs) Approvals	(Completed/ terminated) N	NCT starts (IND file) Sales t Study starts	NCT starts (IND file) Sales $t-1$ Study starts
Log sales	1.3828*	0.8123	0.7056	0.7720	0.7836*	2.2148*	0.8784*
	(0.5444)	(2.7376)	(0.4071)	(1.0532)	(0.3552)	(0.9852)	(0.4877)
First-stage residual	—	−0.7305	−0.6278	−0.8393	−0.6998	−1.9992*	−0.6303*
		(2.7504)	(0.4153)	(1.2342)	(0.3625)	(0.9797)	(0.4434)
$\log(1+N_{m,t-1})$	0.6344	—	—	—	—	—	—
	(0.9301)						
FINAL	Yes	Yes	Yes	Yes	Yes	Excl. publications	Excl. publications
Year effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Class effects	Differenced	Yes	Yes	Yes	Yes	Yes	Yes
Observations	639	630	630	144	630	1227	1227
Classes	71	70	70	18	70	139	139
First-stage observations	—	639	639	639	639	2711	2711
First-stage classes	—	71	71	71	71	311	311
First-stage clustered F	—	0.084	90.989	58.419	58.419	14.657	43.859
Recall-exposed Poisson classes	—	55	9	5	8	30	31
Residual-null test p	—	0.354	0.058	0.490	0.044	0.026	0.251
AR(1) p	0.872	—	—	—	—	—	—
AR(2) p	0.475	—	—	—	—	—	—
Hansen p	—	—	—	—	—	—	—
GMM instruments	21	—	—	—	—	—	—

Notes: 2007–2015. Trial stocks count reported start–completion intervals. All models account for class and year effects. The first five columns use FINAL controls at $t-2$. Standard errors are clustered by class; CF-IV accounts for both estimation stages, and GMM uses the Windmeijer correction. GMM is exactly identified.

Recall intensity is measured per 100 marketed products. The all-recalls column includes audited Class I/II recalls and clinical actions. First-stage F-statistics test the excluded instruments jointly. NME and completed/terminated models retain the preferred sales first stage. Completed/terminated status is recorded on 23 October 2026 and applied retrospectively; termination denotes early stopping.

The NCT-start columns share a broader sample without the publication-coverage restriction, use eight nonpublication controls, and include audited selected Class I recalls plus clinical actions. For current sales, recalls enter at $t-1$ and controls at $t-2$; for lagged sales, they enter at $t-2$ and $t-3$, respectively.

Residual-null tests use covariance estimated under $H_0 : \rho = 0$, so their p-values may differ from coefficient tests. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$; significance levels are unadjusted for multiple comparisons.

[25] See FDA, Investigational New Drug Application.

[26] Our publication measure requires verified ingredient identities and complete PubMed retrieval for every ingredient in a fixed basket of marketed medicines. An unresolved ingredient therefore makes the class-level measure unavailable, even when trial starts, sales and recalls are observed. Imposing this requirement would reduce the common sample from 139 informative markets and 2,045 study starts to 40 markets and 365 starts, excluding several research-intensive oncology and immunology classes. We therefore retain the other eight FINAL controls and market and year fixed effects in this broader-coverage exercise. Market effects absorb persistent class differences, while year effects capture common annual changes. Class-specific changes in scientific opportunities nevertheless remain a potential source of omitted confounding.

Table 9: Market size and innovation: robustness checks, ATC-4.

Outcome	(Two-step GMM) $\log(1+N)$	(Second-stage all recalls) N	(CF-IV) recalls $t,t-1$ N	(NMEs) Approvals	(Completed/ terminated) N
Log sales	1.5893*	0.1833	0.9194*	0.4500	0.9255*
	(0.7788)	(0.1142)	(0.3969)	(0.9914)	(0.3739)
First-stage residual	—	−0.0783	−0.8169*	−0.1223	−0.8172*
		(0.1262)	(0.3973)	(0.9290)	(0.3745)
$\log(1+N_{m,t-1})$	1.1307	—	—	—	—
	(1.1701)				
FINAL	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes
Class effects	Differenced	Yes	Yes	Yes	Yes
Observations	1098	1071	1071	243	1071
Classes	122	119	119	27	119
First-stage observations	—	1098	1098	1098	1098
First-stage classes	—	122	122	122	122
First-stage clustered F	—	6.331	120.578	86.873	86.873
Recall-exposed Poisson classes	—	90	13	6	12
Residual-null test p	—	0.488	0.019	0.895	0.020
AR(1) p	0.563	—	—	—	—
AR(2) p	0.453	—	—	—	—
Hansen p	—	—	—	—	—
GMM instruments	21	—	—	—	—

Notes: 2007–2015. Specification, outcome definitions and inference follow Table 8, using ATC-4 markets. One unresolved NME class–year remains missing. $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$.

The dynamic GMM estimates retain a positive sales coefficient, significant at the 5% level at both aggregations. Their magnitudes are not directly comparable with the Poisson elasticities because the dependent variable is $\log(1+N_{mt})$. Broadening the recall definition yields inconclusive results: the first-stage F-statistics fall to 0.08 at ATC-3 and 6.33 at ATC-4, indicating weak relevance and limiting the reliability of conventional second-stage inference. Adding current recalls to the preferred lagged instrument gives positive sales coefficients of 0.706 and 0.919, although only the ATC-4 estimate is significant at 5%. Greater coefficients at the ATC-4 level may be due to ATC-4's higher granularity relative to ATC-3, implying greater degrees of substitutability.

The NME elasticities are 0.772 at ATC-3 and 0.450 at ATC-4, but both are imprecisely estimated. Few approvals occur in the estimation samples, leaving limited outcome variation despite a strong sales first stage. Repeating the analysis at the broader ATC-2 level also yields a statistically insignificant sales coefficient.[27]

Development timing may also matter. Annual approvals reflect research decisions made years earlier, often in anticipation of future demand (Acemoglu and Linn, 2004). Sunk development expenditure can limit the short-run response to changes in expected revenues (Dubois et al., 2015). Consistent with these timing differences, Finkelstein (2004) finds that vaccine trial entry responds soon after policies raise expected returns, while increases in approvals emerge several years later. The present estimates are too imprecise to determine whether approval flows respond differently from trial activity.

For completed or terminated studies, the estimated elasticities are 0.784 at ATC-3 and 0.926 at ATC-4, both significant at 5%. The positive relationship between sales and trial stocks therefore also appears when attention is restricted to studies subsequently recorded as having ended.

Moreover, the study-start specifications examine whether market rewards affect the initiation of clinical research. With current sales, the estimated elasticity is 2.215 (SE 0.985, $p=0.025$). Under the maintained identifying assumptions, a 1% decline in sales corresponds to approximately 2.2% fewer expected study starts. With sales lagged one year, the elasticity is 0.878 (SE 0.488, $p=0.072$), providing weaker evidence of a positive response. By counting studies in their reported start year, these specifications capture research entry more directly than trial stocks. The positive current-sales estimate is consistent with the timing argument discussed above: changes in expected revenues may affect decisions to initiate studies before their consequences appear in approval flows (Dubois et al., 2015; Finkelstein, 2004). Study starts can also include further research on marketed medicines, an activity that counts of NME approvals do not capture. These differences offer a possible explanation for finding a response in study starts alongside imprecise NME estimates.

[27]Additional checks vary approval coverage, publication matching, sample periods, controls and recall definitions, including a comparison that incorporates new biologics. None of the estimable specifications yields a sales coefficient significant at the nominal 5% level.

Finally, completed/terminated stocks yield sales coefficients of 0.784 (SE 0.355) and 0.925 (SE 0.374), close to the main estimates and significant at the nominal 5% level. The positive relationship documented for the full trial stock also appears in this subgroup.

In the CF-IV columns, the sales coefficient is an elasticity of the expected outcome count under the maintained identifying assumptions. For example, 0.784 corresponds approximately to 7.84% more expected completed/terminated trial stock for a 10% increase in sales. The residual supplies the control-function correction. GMM explains $\log(1+N_{mt})$, so its coefficient describes changes in this transformed outcome. Comparisons across outcomes should account for their different timing and sample coverage. Formal comparisons between coefficients would require joint tests.

7 Conclusions

A long tradition of studies underscores the importance of market size in shaping innovation in the pharmaceutical industry. The feedback between pharmaceutical innovation and market size highlighted by Cerda (2007) makes causal identification a central challenge. Competition and technological opportunities also affect research incentives and must be considered when interpreting the relationship between revenues and innovation (Dubois et al., 2015; Rake, 2017).

This paper contributes to this literature by using major recalls to study how clinical research responds to contractions in therapeutic-class revenues. Combining U.S. sales, recall, and trial data, our preferred control-function IV estimate yields an elasticity of linked clinical-trial stocks with respect to market sales of approximately 0.79. Under the maintained identifying assumptions, a 10% decline in market sales corresponds to approximately 7.9% lower expected trial stock. This interpretation follows the direction of the shocks used for identification. Applying the estimate to revenue increases requires the assumption, imposed by the constant-elasticity specification, that research responds with the same elasticity to rising and falling revenues.

The theoretical framework explains why these revenue changes can matter for research. Firms' development decisions depend on expected returns over the life of a project. A recall may reduce treatment availability or affect confidence in related medicines. If substitution within the therapeutic class and price adjustments do not fully compensate for lost sales, class revenues can fall. A persistent contraction can then reduce the returns expected from research. The evidence that selected recalls predict lower revenues beyond the immediate disruption is consistent with this mechanism and supports the relevance of the instrument. Observed sales serve as a proxy for the expected rewards underlying firms' decisions.

The outcome measured is central to interpreting the estimate. Linked trial stocks count studies over their reported start–completion intervals and can change through both trial entry and study duration. Such activity includes unsuccessful development attempts and research on existing medicines. Estimates using annual new molecular entity approvals are positive but imprecise, so our data provide more informative evidence on research activity than on the arrival of new medicines, implying that comparisons with launch-based estimates, such as those of Dubois et al. (2015), require attention to development lags, attrition and the activities covered by each outcome.

Separating trials by phase provides a further view of this relationship. Research stages differ in their remaining costs, risks and time to potential commercial returns. Among studies with a documented industry sponsor or collaborator, we find positive and statistically significant elasticities for Phases III and IV under both ATC-3 and ATC-4 market definitions. The Phase I and Phase II estimates are imprecise, and the analysis does not establish that later phases are more responsive than earlier ones. The Phase III results link market rewards to advanced clinical testing. The Phase IV findings extend the evidence to postmarketing research on medicines already available to patients, an activity that counts of newly launched drugs do not directly capture.

We implement the recall-based identification strategy using the control-function fixed-effects Poisson approach of Lin and Wooldridge (2019). Market effects account for persistent differences across therapeutic classes, while the control function addresses the endogeneity of sales under the model's assumptions. The causal interpretation requires selected recalls to be unrelated to omitted changes in research conditions and to affect research through the market rewards represented by revenues, conditional on the controls and fixed effects.

Independent effects on development costs, approval prospects or other research conditions remain potential threats to this interpretation.

In order to help explain how these shocks reach the market and the recalling firm, we developed demand and portfolio analyses. A sales loss for one medicine can be partly recovered by competing products or other medicines supplied by the same corporation. Our structural demand model examines substitution among available medicines and the outside option. It compares predicted price responses and substitution under observed recall exposure with a counterfactual that removes the selected recall terms, holding prices and available products fixed. Purchases displaced by a small increase in a generic medicine's price are predicted to move mainly to other generics, with modest changes in diversion towards nongeneric medicines across the two recall states. The portfolio regressions associate higher recall intensity with lower sales in the identified portfolio within therapeutic classes, although estimates for the firm's other products are too imprecise to determine whether they compensate for these losses. Together, these analyses clarify why revenue changes at the product, firm and therapeutic-class levels need to be distinguished when evaluating research incentives.

Our findings are relevant to policies affecting pharmaceutical prices, reimbursement and competition. Changes that reduce expected market rewards may also reduce the research firms are willing to undertake, including work on medicines already available to patients. The policy trade-off therefore includes postmarketing research as well as the development of new treatments. Assessing that trade-off requires evidence on access, treatment quality and health benefits, which trial counts alone cannot provide (Lakdawalla, 2018).

These considerations also bear on the debate over incentives for research in small markets, including orphan drugs and personalized medicine (Garassino et al., 2026; Yin, 2008). Limited expected sales can leave research with substantial potential medical benefits commercially unattractive. Our results reinforce the importance of understanding commercial incentives in these settings, while leaving the size and effectiveness of targeted support to analyses of the markets and policies concerned.

Future research could follow trial starts, continuation and eventual approvals over longer periods to assess how revenue shocks affect the timing and completion of development. Extending the demand model to allow prices and the set of available medicines to adjust would help quantify the broader consequences of recalls. Evidence from more recent data and other sources of revenue variation would also help establish how far the response documented here extends to different market conditions and to expansions in expected rewards.

Appendix

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A Literature review

Table A.1 reviews the relationship between pharmaceutical market size and innovation, starting with Acemoglu and Linn (2004), whose use of demographic variation to address endogeneity provides a reference point for subsequent studies. The review focuses on pharmaceuticals because recall definitions differ across industries. Differences in outcomes and research designs are discussed after the table.

Table A.1: Market size and pharmaceutical innovation: evidence and research designs

Study, sample and data	Unit and outcome	Market size / variation	Estimation	Reported finding
Acemoglu and Linn (2004) US, 1965–2000. March CPS; FDA; OECD.	Broad ATC-2 classes; NMEs and all drug approvals.	Demographic potential market size.	QML.	Elasticity: about 3.5 for NMEs and 6 for all approvals.
Lichtenberg (2006) US, 1952–2001. GLOBOCAN; WHO; IARC; National Library of Medicine; other sources.	Cancer-related MeSH categories; scientific articles and distinct chemotherapy regimens by cancer site.	—	FE.	0.53.
Cerda (2007) US, 1968–1997. FOIA; government R&D funds; US Statistical Abstract.	15 drug categories; NMEs.	Demographic measures.	FE; GLS; IV; Tobit.	Positive initial effect and innovation–mortality feedback; details below.
Blume-Kohout and Sood (2013) US, 1998–2010. Pharmaprojects; MEPS; OECD; NIH.	49 therapeutic classes; development activity from preclinical research through Phase III.	Medicare Part D; demographic exposure.	Negative binomial; Poisson.	Phase I increases: 26%, 33%, 51%; implied elasticity: 2.4–4.7. Periods below.
Dubois et al. (2015) 14 countries, 1997–2007. IMS; WHO.	Chemical entities; NCEs, with ATC-1 and ATC-2 indicators.	Sales revenues; deaths and GDP as instruments.	OLS; 2SLS; CF (Wooldridge, 2002).	Elasticity: 0.23 in the preferred specification.
Rake (2017) US, 1974–2008. R&D; FDA; OECD. ClinicalTrials.gov.	Disease classes; NME and NDA approvals; Phase II and III trials.	Demographic measures.	Entry-stock Poisson QMLE (Blundell et al., 1995).	0.3444 (NME); 0.3521 (NDA).

Agarwal and Gaule (2022) US, 2015–2020. GBD. ClinicalTrials.gov.	75 indications; clinical trials.	—	Poisson.	0.43.

Notes: Findings refer to different outcomes and specifications and are not directly comparable. A dash denotes a market-size proxy not specified in this summary. Study-specific construction and interpretation are given below.

Abbreviations: ATC, Anatomical Therapeutic Chemical classification; CF, control function; CPS, Current Population Survey; FDA, Food and Drug Administration; FE, fixed effects; GLS, generalized least squares; IMS, IMS Health; IV, instrumental variables; MEPS, Medical Expenditure Panel Survey; NCE, new chemical entity; NME, new molecular entity; NDA, new drug application (approvals are counted here); QML/QMLE, quasi-maximum likelihood (estimation/estimator); 2SLS, two-stage least squares; WHO, World Health Organization.

Interpretation and comparability. Blume-Kohout and Sood (2013) measure development activity, whereas Acemoglu and Linn (2004) and Dubois et al. (2015) examine approved or marketed drugs. Demographic changes and the Medicare Part D reform also provide different sources of variation; Dubois et al. (2015) recognise the latter identification strategy. Blume-Kohout and Sood (2013) further argue that price regulation and changing rules in several countries in Dubois et al.'s sample may lower expected profits per consumer and increase uncertainty, weakening R&D responses to a given revenue change relative to the US (Sood et al., 2009).

Medicare Part D evidence. For a class with the average Medicare market share of 41% in 2004–2005, Duggan and Scott Morton's revenue estimate implies an 11% increase after Part D. For the same exposure, Blume-Kohout and Sood (2013) report Phase I increases of 26% in 2004–2005, 33% in 2006–2007 and 51% in 2008–2010, implying elasticities of 2.4–4.7. These bracket Acemoglu and Linn's approximately 3.5 for NMEs. Combining all trials, including Phase III studies of supplemental indications, yields a lower elasticity than their approximately 6 for all drug approvals, but a higher one than Dubois et al.'s approximately 0.25 (0.23 in the table). This evidence links expanded outpatient prescription coverage to greater pharmaceutical R&D.

Data and outcome construction.

Cerda: government pharmaceutical R&D funds measure research inputs; Statistical Abstract population data measure market size. Cerda reports approximately 0.08 additional drugs following an exogenous market expansion, lowering mortality among people aged 65 or older by 0.8%; the resulting demand feedback produces a further 0.096 drugs.

Rake: following Cerda's drug–indication matching approach, the 19th edition of the *Drug Information Handbook* (Lexi-Comp/American Pharmaceutical Association; Lacy et al., 2010) supplies active ingredients, indications and other information, including adverse effects. Only FDA-labelled indications are included, excluding unlabelled and investigational uses. The 1974–2008 data contain 599 unique NMEs and 1,665 unique NDA approvals across 208 indications; a drug is counted in every indication for which it is approved.

Dubois et al.: IMS sales data cover Australia, Brazil, Canada, China, France, Germany, Italy, Japan, Mexico, Korea, Spain, Turkey, the UK and the US. The database contains 607 ATC-4 classes, 6,216 active ingredients and 471 formats, plus producer, licensing and patent-start information. The authors describe ATC-4 classes as sharing indications and mechanisms of action. OTC products are excluded; IMS standard units represent the smallest typical dose of a product form.

Blume-Kohout and Sood: Pharmaprojects trend snapshots cover preclinical development through Phase III, rather than using ClinicalTrials.gov. This captures one component of innovation, as discussed by Hall and Rosenberg (2010).

Controls.

Acemoglu and Linn: scientific and other supply-side incentives; lagged innovation; life-years lost; public funding; pre-existing sector and major-category trends; and health-insurance market size (pp. 1077–1080).

Cerda: Medicare and Social Security spending; research grants; year effects; disease-specific hospital-use composition/prevalence by sex, race and marital status; and patients' mean age.

Rake: indication-year knowledge stock from Web of Science publications linked to BioPharmInsight indications; regulatory stringency measured by FDA submission-to-approval time; pre-sample mean innovation; indication-specific mortality in 1983; and technological opportunities measured by average annual knowledge-stock growth in 1979–1983. These capture demand-pull and technology-push factors.

Blume-Kohout and Sood: prescription-drug measures and disease-class research grants.

B Recall data: collection and attribution

This appendix explains how we assembled the recall catalog and linked it to the US pharmaceutical sales data. We distinguish three steps: collecting a documented action, identifying the corresponding product and corporation, and determining its eligibility for a particular instrument definition. Inclusion in the catalog does not by itself establish a market-wide effect or instrument validity.

B.1 Sources and coverage

The main sources are FDA notices, openFDA, withdrawal reviews by Onakpoya et al. (2016), the WITHDRAWN database of Siramshetty et al. (2016), the WHO Collaborating Centre for Drug Statistics Methodology website, and PubMed. Historical lists and the accompanying

source spreadsheets provided candidate names, final action dates, reasons and scope were checked against regulatory documents or original company announcements.[B.1] Worldwide lists were not treated as lists of US events. We checked whether each action applied in the United States and distinguished safety or efficacy actions from commercial exits, indication changes and formulation-specific restrictions.

The historical FDA extract was obtained through the Freedom of Information Act (FOIA). The supplied workbook is identified as *FDA FOIA Response 2020-2770* and contains Class I and Class II product-recall records classified from January 2004 to December 2012. Economic timing follows the recall initiation date, rather than the classification date, publication date or year embedded in a recall number. Clinical actions are dated using the documented announcement or marketing cessation.

We supplemented this extract with the openFDA drug enforcement records and the FDA enforcement export. For each record, we retained the FDA event identifier, recall number, description, recalling firm, reason, classification, dates and available distribution or lot information. Records appearing in several sources were reconciled by recall number and reviewed at the event level. A later notice or newly recovered product record therefore does not automatically create a new recall event.

The supplementary search covers actions dated 2004–2015, while the first-stage analysis uses market histories over 2004–2013. The preferred one-year-lag specification has outcome years 2007–2013 and thus uses actions recorded in 2006–2012. Table B.1 lists supplementary clinical actions retained in the source catalog and additional Class I notices covered by at least one of the three candidate cause definitions. Other supplementary enforcement records remain in the inventory without being individually listed in this focused table and are available upon request.

B.2 Identifying products and corporations

We followed strict rules to map the recalls from openFDA to our datasets. The reason for this is that our dataset takes a snapshot of the products in our dataset in the second quarter of 2015. The recalls on openFDA, however, refer to the product's historical information at the time of the recall. This means that information like mergers and acquisitions, acquisitions of product lines, division of firms, or selling of product lines should be manually assessed in order to properly attribute the recall to the correct manufacturer. Furthermore, when a distributor issues a recall, the supply chain should be reconstructed to identify the original manufacturer, since our dataset includes wholesalers. Manufacturers, distributors, and repackagers may, moreover, report the same medicine under different names or identifiers. We follow three steps.

1. *Recover product identifiers.* We extract reported National Drug Codes (NDCs), retain their original representation, and reconcile formatting differences with FDA product listings, historical or obsolete listings, NIH RxNav and DailyMed records. Notices listing several NDCs are expanded into recall–NDC pairs for matching. This expansion identifies packages.

2. *Resolve the firm behind the product.* An NDC identifies a labeler, product and package. The labeler can be a manufacturer, distributor or repackager, so its identity is not automatic proof of manufacturing ownership. We check the notice, product and application records, historical labels and documented corporate relationships. Manufacturer abbreviations, particularly in Advantage Dose notices, are used only when corroborated by the product description, an NDC or other supporting evidence.

3. *Match to the sales panel.* We harmonize product and corporation names and check available ingredients, formulation, strength, route and prescription/OTC status. The final match must correspond to a product–corporation pair represented in the sales data. Unresolved names, conflicting identities and flagged scope problems remain distinguishable from confirmed matches. The absence of a match is not interpreted as evidence that no recall occurred.

[B.1] The withdrawal reviews include Onakpoya et al., *Critical Reviews in Toxicology* (2016) and Onakpoya et al., *BMC Medicine* (2016). The WITHDRAWN database includes several reasons for discontinuation; a database entry alone does not establish a US safety withdrawal.

We manually reviewed mergers, acquisitions, company divisions and transfers of individual product lines to reconcile historical recall notices with the corporate structure represented in the sales panel. Where the panel consolidates a subsidiary's products under its parent, the corresponding recalls were assigned to that parent. Later ownership changes were not automatically applied to firms still recorded separately in the data. Acquisitions or divestitures involving only part of a portfolio were assessed product by product, using historical NDCs, archived labels, transaction announcements and company filings. This assessment determined which firm–product pair in the panel corresponded to the recalled medicine. Where the scope or timing of a transfer remained uncertain, we documented the case-specific attribution assumptions. Since until 2010 it was common practice in the U.S. that retailers or distributors issuing a recall replace their NDC with the NDC of the manufacturer, for notices carrying a repackager's NDC, we also signed and submitted a targeted FDA FOIA request seeking the original manufacturer NDCs and underlying source-product information. As an example, the supplementary Heartland Repack Services crosswalk identifies manufacturer names and application numbers associated with repackaged products. We combined this information with recall-specific evidence[B.2] The role of NDC evidence follows the FDA's description of the NDC system.

B.3 Scope checks and construction of recall exposure

Table B.1 documents additional withdrawals and other (publicly available) clinical safety or efficacy actions identified from withdrawal reviews and databases. We checked each candidate against FDA documents or original company announcements, recording its applicability to the United States, the affected medicines, the clinical reason, the action date, and its stated scope. These checks distinguish complete withdrawals from temporary suspensions and actions limited to particular formulations or uses. An action affecting one supplier was extended to other versions of the medicine only when the documentation supported that scope. Commercial discontinuations were recorded separately, and we checked whether an action was already represented in the FDA files to avoid duplication. If the discontinuation was not preceded by an adverse event, we did not consider it among our set of recalls. The table also reports additional Class I notices recovered from FDA enforcement reports that were not present in openFDA.

These additions extend the catalog beyond batch-specific recalls to actions that can remove marketed treatments from use or change perceptions of their safety or effectiveness. Their relevance to studying market size follows from the importance of potential market rewards for pharmaceutical innovation (Acemoglu and Linn, 2004) and evidence that safety information can change physicians' prescribing choices (Dubois and Tunçel, 2021). Whether competing medicines replace the affected sales, leaving total class revenues unchanged, remains an empirical question.

Table B.1: Additional actions covered by the candidate recall definitions

Year	Medicine	Documented action or reason	Source
A. Supplementary clinical safety or efficacy actions			
2004	Vioxx (rofecoxib)	Worldwide withdrawal following cardiovascular trial findings.	Merck/SEC
2005	Cylert and generic pemoline	US marketing ceased because of liver-toxicity concerns.	FDA history
2005	Tysabri (natalizumab)	Temporary US withdrawal following progressive multifocal leukoencephalopathy (PML) reports.	FDA history
2005	Bextra (valdecoxib)	Withdrawal following cardiovascular concerns and serious skin reactions.	FDA review
2007	Permax and generic pergolide	US withdrawal covering branded and generic human medicines.	FDA history
2007	Zelnorm (tegaserod)	General US marketing suspended following cardiovascular ischemic-event concerns.	FDA history
2007	Trasylol (aprotinin)	Marketing suspended following a trial mortality signal.	FDA history
2009	Raptiva (efalizumab)	Phased US withdrawal following PML cases.	FDA release

Continued on next page

[B.2]The FOIA agreement was signed in October 2025.

Table continued

Year	Medicine	Documented action or reason	Source
2010	Mylotarg (gemtuzumab ozogamicin)	Withdrawal following lack of confirmed benefit and safety concerns at the original regimen.	FDA release
2010	Meridia (sibutramine)	US withdrawal following cardiovascular trial findings.	FDA release
2010	Propoxyphene medicines	US withdrawal of Darvon, Darvocet and generic single-ingredient and combination products.	FDA release
2011	Xigris (drotrecogin alfa)	Worldwide withdrawal following failure to demonstrate a survival benefit.	Lilly
2013	Omontys (peginesatide)	All marketed lots recalled following serious or fatal anaphylaxis reports.	Takeda
B. Additional Class I enforcement notices			
2012	Morphine cartridges	Overfilled cartridges; D-1419-2012.	FDA 61624
2012	Hydromorphone cartridges	Overfilled cartridges; D-1444-2012.	FDA 61847
2012	Hydromorphone cartridges	Overfilled cartridges; D-020-2013, initiated in 2012.	FDA 62854
2013	Mitosol kits	Yeast contamination in kit components; D-809-2013.	FDA 64018
2013	Piperacillin/tazobactam	Precipitation or crystallization on reconstitution; D-806-2013. Broader quality definition only.	FDA 64657
2013	Warfarin tablets	Oversized tablets; D-838-2013.	FDA 65394

Notes: This table lists additional retained candidate actions, not all supplementary source records or all observations in the instrument. Clinical actions need not carry a Class I designation. Years refer to initiation or the documented action; source documents may have been published later. The preferred lagged instrument uses action years 2006–2012, so entries from 2004–2005 and 2013 are outside that exposure window. A listed action does not establish a market-wide contraction, permanent withdrawal, manufacturing defect, or exclusion validity.

In order to provide evidence on the representativeness of our recalls, we compare the collected catalog with published recall counts before matching products to sales or selecting an instrument. A product recall notice is distinct from a recall event: a single event can generate multiple product recall notices. The comparable benchmark uses Class I and Class II CDER products classified in each FDA fiscal year (October–September). The catalog is counted on the same basis, using distinct FDA event–recall-number pairs. Aggregated source entries are expanded to individual notices. The D prefix identifies CDER records, following the FDA recall-number convention. Compounding and repackaging notices are retained, while Class III notices and clinical actions without an FDA classification are excluded from the collected comparison series. We also report counts by recorded action year. The action year and the fiscal classification year can differ because classification occurs later and fiscal years begin in October. Economic analyses continue to use action timing. For 2009, the FDA's all-class total is shown separately because its Class II/III split is unavailable.

Table B.2: Published annual recall counts and the comparable benchmark

	RAPS I+II	Hall et al. I–III	FDA, FY I–III	Comparable mean I+II, FY	catalog I+II Action year	catalog I+II Fiscal year
2004	79	—	318	184	142	153
2005	148	—	502	332	370	332
2006	110	—	361	234	570	234
2007	67	—	988	881	489	878
2008	146	—	379	212	1,751	212
2009	108	—	1,984	—	525	1,851
2010	170	—	868	547	638	546
2011	302	—	1,616	1,370	2,205	1,370
2012	410	418†	1,703	1,572	568	1,572
2013	1,090	1,283	1,065	910	1,915	910
2014	741†	1,344	1,647	1,498	1,180	1,498

Notes: RAPS reports FDA-derived entries available on 11 August 2014; the underlying year field and historical completeness are not established. Hall et al. use calendar report dates and include biologics and dietary supplements; an annual Class III breakdown is unavailable. †Partial year: Hall starts on 20 June 2012; RAPS ends on 11 August 2014. Neither series enters the comparable mean. FDA links identify fiscal-year product charts; the Class I–II benchmark adds the two classes. For 2014, the linked official openFDA query supplies the breakdown (153 Class I, 1,345 Class II, 149 Class III), whose sum matches the published all-class total. Only one comparable FDA source series enters each nonmissing mean ($n = 1$); repeated FDA-based sources are not counted as independent estimates. For 2009, 1,984 is a verified FDA total including Class III; the Class I–II mean and ratio remain unavailable. A dash denotes an unavailable comparable count, not absence of collected recalls. Source definitions, query dates and exclusions are preserved in the source ledger.

Figure B.1 shows close count agreement in years with a comparable benchmark from 2005 onward. In 2004 the catalog contains 153 notices against 184 in the published benchmark; the differences in 2007 and 2010 are three and one notices, respectively. Agreement in aggregate counts is a coverage check, not proof that every record is captured or that the matched analytical sample is representative. The catalog and the published statistics also share FDA source records.

The source catalog contains 527 records with action year 2009: 525 Class I–II product notices and 2 additional clinical actions without an FDA classification. Separately, 1,851 Class I–II notices were classified in fiscal 2009, including actions initiated in other calendar years. These are different timing conventions, rather than missing observations.

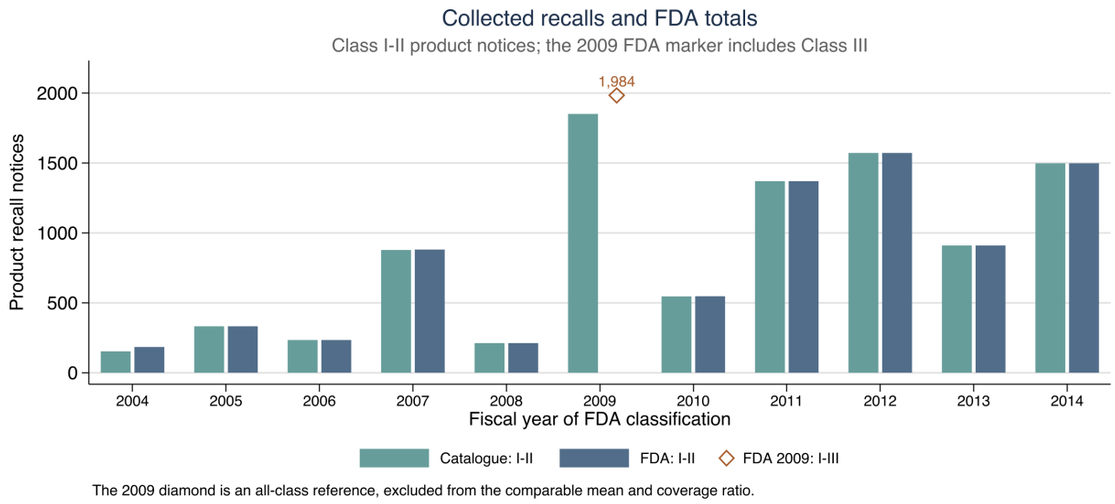


Figure B.1: Collected product recall notices and FDA reference counts, by fiscal year of classification. Bars compare collected Class I–II notices, before sales matching and instrument selection, with the comparable benchmark in Table B.2. The separate diamond in 2009 shows the verified FDA total of 1,984 notices including Class III; it is excluded from the Class I–II mean and coverage ratio. Compounding and repackaging notices are retained. Counts refer to product notices, rather than independent recall events.

C Control-function fixed-effects Poisson estimation

This appendix supplements the model in Equation (6) and the estimation procedure in Equations (9)–(10). Namely, we provide the additional conditional-mean restriction underlying the control-function correction and explain its connection to the Mundlak representation, following Lin and Wooldridge (2019). The economic rationale and identifying assumptions for the recall instrument are discussed in Section 5.1.

Throughout this appendix, m indexes therapeutic markets, $M_{mt} = \log(\text{Sales}_{mt})$, as in Equation (4), and z_{mt} contains the FINAL controls and year effects, with coefficients γ_c. Let $r_{mt} \equiv \widetilde{R}_{m,t-1}$ denote the excluded instrument in the preferred specification.

C.1 The control-function restriction

Let $\mathcal{W}_m = \{z_{ms}, r_{ms} : s = 1, \ldots, T\}$ denote the complete control and instrument history, and let $\mathbf{u}_{m2} = (u_{m12}, \ldots, u_{mT2})'$ collect the first-stage disturbances. Maintain the structural mean in Equation (6) conditional on the full regressor history and the first-stage market effect c_{m2}. The linear reduced form is assumed correctly specified, with

$$\mathbb{E}[u_{mt2} \mid \mathcal{W}_m, c_{m2}] = 0.$$

A sufficient control-function restriction is

$$\mathbb{E}[\exp(\kappa_{mt}) \mid \mathcal{W}_m, \mathbf{u}_{m2}, c_m, c_{m2}] = \exp(\psi + \rho u_{mt2}), \tag{C.1}$$

where ψ and ρ are constants. This restricts the conditional expectation of $\exp(\kappa_{mt})$, rather than merely the conditional mean of κ_{mt}. It is an additional functional-form and exogeneity assumption; instrument relevance alone does not imply it.

Because the reduced form determines M_{mt} from the variables in this conditioning set, iterated expectations give

$$\mathbb{E}[N_{mt} \mid \mathcal{W}_m, \mathbf{u}_{m2}, c_m, c_{m2}] = c_m \exp\Big(\psi + \beta_1 M_{mt} + z'_{mt}\gamma_c + \rho u_{mt2}\Big). \tag{C.2}$$

The first-stage disturbance therefore enters the exponential mean as an additional regressor.

C.2 Within residuals and the Mundlak representation

Write

$$u_{mt2} = \bar{u}_{m2} + \ddot{u}_{mt2}, \qquad \bar{u}_{m2} = \frac{1}{T}\sum_{t=1}^{T} u_{mt2}.$$

Equation (C.2) can then be expressed as

$$\mathbb{E}[N_{mt} \mid \mathcal{W}_m, \mathbf{u}_{m2}, c_m, c_{m2}] = g_m \exp\Big(\beta_1 M_{mt} + z'_{mt}\gamma_c + \rho\ddot{u}_{mt2}\Big), \qquad g_m = c_m \exp(\psi + \rho\bar{u}_{m2}). \tag{C.3}$$

Since g_m is constant over time within a market, it is absorbed by the ATC fixed effects. This explains why the estimated within residual used in Equation (10) can replace the unobserved first-stage disturbance. It does not require $\bar{u}_{m2} = 0$.

The same result can be expressed using Mundlak residuals. Define $w_{mt} = (z'_{mt}, r_{mt})'$ and $\bar{w}_m = T^{-1}\sum_t w_{mt}$. Consider the pooled first-stage regression

$$M_{mt} = \theta + w'_{mt}\Pi_w + \bar{w}'_m\Phi_w + v_{mt2}, \tag{C.4}$$

including the time averages of all first-stage regressors. Its OLS residuals satisfy

$$\widehat{v}_{mt2} - \overline{\widehat{v}}_{m2} = \ddot{\widehat{u}}_{mt2}.$$

Thus, the Mundlak and fixed-effects residuals differ only by a market-specific constant. In an FE Poisson model, that constant is absorbed by the market effect, yielding identical slope estimates with either residual. No additive specification for the multiplicative outcome effect c_m is required.

Under these conditional-mean restrictions and standard regularity conditions, FE Poisson QMLE consistently estimates the slope parameters without requiring a Poisson conditional variance or serial independence. Our standard errors cluster by market and account for first-stage estimation uncertainty. Because M_{mt} is log sales, β_1 is an elasticity of the conditional expected trial count with respect to sales, under the maintained identification assumptions discussed in Section 5.1.

D Structural model with exogenous recalls

We study substitution among medicines and the outside option using a random-coefficients demand model following Nevo (2000) and the pharmaceutical applications of Dubois and Lasio (2018) and Dubois et al. (2022). The model allows recalls to affect the attractiveness of medicines and their price sensitivity. The broad idea is to construct a model similar to that of Dubois et al. (2018), where – rather than building counterfactuals on a ban on advertising from a structural model considering how consumers react to advertising – we compare price elasticities under observed recall exposure with those obtained after setting the selected recall exposures to zero. In our application, advertising is substituted by exogenous recalls.

To study substitution among medicines, we use annual U.S. retail data for 2004–2015, which record purchases shaped by the combined decisions of patients, prescribers, pharmacists, and insurers. We represent the alternatives available to these decision-makers as products distinguished by firm, international, and local product names, ingredient combinations, ATC-4 class, form, and strength. Within each product, we aggregate pack variants and calculate price as annual revenue per standard unit. We then group these products into choice sets defined by ATC-3 therapeutic classes, aligning the demand analysis with the main analysis and allowing substitution across the ATC-4 categories within each class. Sensitivity checks at ATC-4 level are provided. Importantly, within each class-year, our counterfactual holds the observed product set fixed. We compare price elasticities and substitution patterns under observed recall exposure and after removing the selected recall terms. This comparison is useful because recalls may alter buyers' valuations of medicines that remain on sale, redistributing purchases and revenues within the class. Positive annual sales may nevertheless include partial or temporary shortages. In other words, our model does not currently account for the fact that complete withdrawals change patients' choice sets, as this would require a more complex structural model, which is beyond the scope of the present work and is left for future research.

Let j index products, m therapeutic classes, and t years, and let J_{mt} denote the products offered in a class-year. We measure exposure to a product's own recalls and to recalls affecting other products. The own-recall state C_{jmt} counts distinct selected issue families affecting product j. The selected events are the major recalls in the main analysis, excluding withdrawals. As mentioned, each family is assumed to remain active in the recall year and the following year while shortages are allowed. A value of $C_{jmt} = 0$ records the absence of documented exposure to this selected event set during the relevant exposure window.

For an individual i, utility from product j is

$$\begin{aligned} u_{ijmt} &= \eta_j + \ell_t + \xi_{jmt} + \alpha \log p_{jmt} + \gamma C_{jmt} \log p_{jmt} \\ &\quad + \lambda_i C_{jmt} + \rho_i R_{jmt} + \kappa_H H_{jmt} + \kappa_{HC} H_{jmt} C_{jmt} + \epsilon_{ijmt}, \\ \lambda_i &= \bar{\lambda} + \sigma_C \nu_{Ci}, \qquad \rho_i = \bar{\rho} + \sigma_R \nu_{Ri}, \end{aligned} \tag{D.1}$$

where

$$R_{jmt} = \sum_{\substack{k \in J_{mt} \\ k \neq j}} C_{kmt}.$$

denotes the exposure to other products' recalls. This measure includes other products of the same firm. A family affecting several presentations contributes once for each affected presentation, so R_{jmt} aggregates exposure across other offered products. Product effects η_j absorb persistent characteristics, including the main effects of form and generic type, while year effects ℓ_t capture common annual changes. The term ξ_{jmt} represents remaining unobserved demand factors. The taste draws ν_{Ci} and ν_{Ri} are independent standard normals, and ϵ_{ijmt} is an independent type-I extreme-value error. The indicator H_{jmt} records whether a hospital source price is observed. Its main effect and interaction with C_{jmt} account for differences associated with source-price coverage.

The specification allows own recalls to change both product attractiveness and price sensitivity:

$$\frac{\partial u_{ijmt}}{\partial p_{jmt}} = \frac{\alpha + \gamma C_{jmt}}{p_{jmt}}, while$$

the coefficients λ_i and ρ_i allow individuals to differ in their responses to own- and other-product recalls (Dubois et al., 2018). Although estimation relies on aggregate product shares, variation in purchases across products and years can provide information about the distribution of these

responses, under the model's distributional and identification assumptions. Since individual purchases and demographic characteristics are unavailable, this heterogeneity remains latent and cannot be attributed to specific consumer groups. We assume that recall responses are common across pharmaceutical forms.

We use log price to accommodate the wide price range across medicines. The price coefficient α and the recall–price interaction γ are common across individuals. We retain random coefficients for the two recall variables [D.1].

Writing $V_{ijmt} = u_{ijmt} - \epsilon_{ijmt}$ and normalizing systematic outside utility to zero gives the individual choice probabilities and aggregate shares (Nevo, 2000):

$$P_{ijmt} = \frac{\exp(V_{ijmt})}{1 + \sum_{k \in J_{mt}} \exp(V_{ikmt})}, \qquad s_{jmt} = \int P_{ijmt}\, d\Phi_2 = \frac{q_{jmt}}{\widehat{\mathcal{M}}_{mt}}, \tag{D.2}$$

where Φ_2 is the joint distribution of the two independent normal taste draws. The potential market size $\widehat{\mathcal{M}}_{mt}$ measures standard-unit purchase opportunities. It determines how observed product quantities are converted into shares and allows some purchases to be allocated to the outside option.

We calibrate the unobserved outside quantities using the annual auxiliary-logit procedure of Huang and Rojas (2013, 2014), adapted from the implementation in Dubois and Lasio (2018). First, a target 2SLS regression uses log product quantity relative to the quantity of the largest inside product in each class-year. Controls and instruments remain undifferenced. Second, annual outside quantities q_{0mt} are chosen to minimize the sum of squared differences between the named target coefficients – computed first from observed inside-product quantities – and the corresponding coefficients from a regression with $\log q_{jmt} - \log q_{0mt}$, as the dependent variable using the same auxiliary regression operator. The calibrated quantities imply

$$\widehat{\mathcal{M}}_{mt} = \sum_{j \in J_{mt}} q_{jmt} + \widehat{q}_{0mt}, \qquad s_{0mt} = 1 - \frac{\sum_{j \in J_{mt}} q_{jmt}}{\widehat{\mathcal{M}}_{mt}}.$$

These outside shares depend on the calibration and its maintained auxiliary specification. Several outside-quantity paths may match the same coefficients.[D.2] We use a fixed auxiliary specification for this calibration. The expanded instrument set described below enters the final demand estimation.

Table D.1 reports the outside-share calibrations for eight ATC-4 classes and compares them with the U.S. hospital shares reported by Dubois et al. (2022). Our hospital calibration is included as an additional comparison.

Table D.1: Comparison of outside-good market shares

ATC-4	Dubois et al. U.S. hospital (%)	Our retail calibration (%)	Our hospital calibration (%)
C2A2	14.0	8.2	10.9
C8A0	13.0	11.1	12.1
L1B0	13.0	24.5	15.9
L4X0	14.0	15.1	17.4
N1A2	14.0	20.3	15.7
N1B1	15.0	9.5	8.3
N5B3	10.0	18.9	15.5
N6A4	16.0	8.8	8.5
Mean	**13.6**	**14.5**	**13.0**

Notes: Entries are outside-good shares expressed as percentages. The first column reports U.S. hospital shares from Dubois et al. (2022); the remaining columns report our auxiliary retail and hospital calibrations for the same eight ATC-4 classes. The retail calibration enters the demand model estimated here. The comparison is descriptive; the calibrated shares retain their dependence on the auxiliary specification.

[D.1] Exploratory specifications with price and non-generic taste dispersion yielded zero or uninformative estimates.

[D.2] With the same auxiliary regression operator, setting the outside quantity equal to the reference-product quantity is an exact matching candidate whenever it satisfies the lower bound. Other outside-quantity paths may yield the same matched coefficients.

Differences across columns may reflect the sales channel, sample period and auxiliary specification. The comparison study covers 2002–2008, and our data cover 2004–2015.

The calibrated shares also provide a direct connection to the revenue aggregation used in Section 3. For each medicine, $q_{jmt} = \widehat{\mathcal{M}}_{mt} s_{jmt}$. Summing across medicines gives total retail purchases within the therapeutic class:

$$Q_{mt}^{in} = \sum_{j \in J_{mt}} q_{jmt} = \widehat{\mathcal{M}}_{mt}(1 - s_{0mt}).$$

Multiplying each quantity by its price and summing gives total retail revenue:

$$\text{Sales}_{mt} = \sum_{j \in J_{mt}} p_{jmt} q_{jmt} = \widehat{\mathcal{M}}_{mt} \sum_{j \in J_{mt}} p_{jmt} s_{jmt}.$$

Its logarithm, $M_{mt} = \log(\text{Sales}_{mt})$, is the corresponding revenue-based measure of market size for the retail market. Notice that $\widehat{\mathcal{M}}_{mt}$ counts potential purchase opportunities, while M_{mt} measures log revenue. Section 3 uses the same principle of aggregating product revenues [D.3]. Estimating price responses requires instruments because both $\log p_{jmt}$ and $C_{jmt} \log p_{jmt}$ may be correlated with ξ_{jmt}. Following the source-price logic of Dubois and Lasio (2018), we construct a hospital log-price residual by removing molecule, year and channel effects from pooled retail and hospital observations. When a source residual is missing, we use the molecule-year mean, followed by the year mean where necessary. We also use five product-composition instruments: counts of rival generic and nongeneric products within the market, the corresponding two counts within ingredient/form, and the number of other products offered by the same firm. Rival counts comprise products supplied by other firms, and the same-firm count is constructed separately.

These six variables, their six interactions with C_{jmt}, and the two additional variables $H_{jmt} p_{jmt}^H$ and $C_{jmt} H_{jmt} p_{jmt}^H$ provide fourteen excluded instruments, where p_{jmt}^H is the observed hospital price in levels and H_{jmt} indicates whether a hospital source price is available for product j in class m and year t. We construct the last two instruments using observed hospital prices and set them to zero when the source price is unavailable. The terms H_{jmt} and $H_{jmt} C_{jmt}$ enter utility as controls for source availability.

We face three identification challenges: distinguishing recall effects from other changes in demand, identifying price responses, and identifying heterogeneity in consumers' recall responses.

First, our recall counterfactual asks how purchases would differ without the selected recall exposures, holding prices and the offered product set fixed. Recall exposure is measured from documented events; its effects are estimated using changes in exposure across product-years and the corresponding purchase patterns. Product effects account for persistent differences in attractiveness, while year effects account for common annual changes. Our institutional premise is that the selected severe recalls become unavoidable once a defect is detected. A causal interpretation additionally requires own- and other-product recall exposure to be unrelated to remaining demand shocks after these controls. Other events may occur, but they must not systematically confound the recall comparison. For example, production problems could cause both a recall and other sales disruptions. Separating own- and other-product responses also requires changes in total class recall exposure to be unrelated to omitted class-specific demand changes.

Second, prices may respond to unobserved demand. We therefore use variation predicted by the pharmaceutical instruments described above to estimate the responses to log price and its recall interaction. Hospital prices are intended to capture shared cost and pricing components, while product-composition instruments reflect competitive conditions. These variables must predict the endogenous price terms without being correlated with remaining unobserved demand. Their validity must also hold across recall states to justify the recall-interacted instruments.

Third, we allow own- and other-product recall responses to follow two independent normal distributions. Information about their dispersions comes from aggregate purchase patterns under different prices and recall exposures. Unlike Dubois et al. (2018), we do not observe repeated household purchases or household-specific exposure. Identification therefore relies on the stated distributional restrictions and on sufficiently informative aggregate variation.

[D.3] As mentioned, the revenue total defined here covers the retail channel.

We estimate the model by one-stage GMM using aggregate shares. For each candidate distribution of recall responses, share inversion recovers mean utilities that match the observed shares. The GMM moments then impose orthogonality between the remaining demand error ξ_{jmt}, the included exogenous variables, and the excluded instruments Nevo (2000) [D.4]. Covariance estimates are clustered by therapeutic class.

As mentioned above, identification requires selected recall exposure to be unrelated to remaining demand shocks, conditional on the controls. The excluded instruments must predict prices and be uncorrelated with these shocks.

We calculate price elasticities conditional on the observed recall state. Defining $a_{kmt} = \alpha + \gamma C_{kmt}$, the elasticity of demand for product j with respect to the price of product k is

$$\varepsilon_{jk,mt} = \frac{a_{kmt}}{s_{jmt}} \int P_{ijmt} \left[\mathbf{1}\{j = k\} - P_{ikmt} \right] d\Phi_2. \tag{D.3}$$

Own-price elasticities set $j = k$. Reported mean cross-price elasticities average over other offered products in the same class-year. Product-level summaries use equal weights across the same observed years in both states, with recall-year summaries reported separately.

For the no-recall comparison, we set $C_{jmt} = R_{jmt} = 0$, removing their interactions and random-coefficient loadings.

This change in the recall state parallels the change in advertising exposure in Dubois et al. (2018). Its scope is the response of demand within the observed set of products at observed prices. Any changes in predicted class revenue arise from changes in purchases of the available medicines, including substitution across products and the outside option. The reported counterfactual results consist of conditional point comparisons of elasticities.

Figure D.1 shows how strongly purchases of the 115 selected recalled generic products respond to their own prices. The horizontal axis reports the approximate percentage quantity loss following a hypothetical 1% price increase.

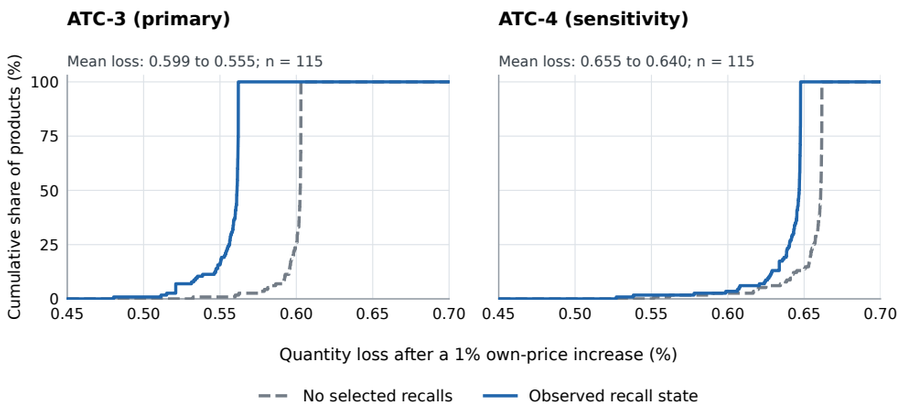


Figure D.1: Own-price responsiveness of selected recalled generics. Curves show the percentage of the 115 products whose predicted quantity loss after a hypothetical 1% own-price increase is at or below each horizontal-axis value. Solid blue denotes observed recall exposure; dashed grey removes the selected recall terms. Both states use the same baseline prices, offered products, fitted unobservables and calibrated market sizes.

The vertical axis gives the percentage of products whose predicted loss is at or below that value. Both curves describe the same products: blue evaluates their (predicted via the random coefficient model above) responses under observed recall exposure, while grey removes the selected recall terms and represents the counterfactual responses. The two states use the same baseline prices, offered products, fitted unobservables, and calibrated market sizes. Intuitively, a curve further to the left therefore indicates lower price responsiveness. In ATC-3, the average predicted quantity loss is approximately 0.56% in the observed state and 0.60% without selected recalls; ATC-4 shows a smaller difference in the same direction. ATC-4 groups are narrower and may contain more similar medicines; however, they also

[D.4]The inversion step adjusts products' average attractiveness so that the model reproduces observed market shares. Different distributions of consumer recall responses can therefore reproduce the same shares. A good match alone does not show which distribution is supported by the data or whether the estimated effects are causal; these conclusions also require informative variation and the identifying assumptions discussed above.

exclude alternatives in neighboring ATC-4 categories that remain available in the ATC-3 model. The fact that price responsiveness changes less under the two-recalls specification may indicate that the former effect prevails.

This Appendix has shown that controlling for competition measures is necessary in our context, where a recall may change substitution patterns between drugs.

E Selected Recalls and Potential Within-Firm Sales Offsets

Sales losses on recalled medicines may be partly offset if purchases shift towards other products marketed by the same firm. Such compensation would limit the firm's overall revenue loss and, when it occurs within the same therapeutic class, could also reduce the contraction in class revenues. Appendix D examines model-predicted substitution patterns under different recall states. This appendix complements that analysis by examining whether lower observed sales in the portfolio linked to selected recalls are accompanied by higher sales elsewhere within the same corporation.

We distinguish sales of a fixed group of product–firm pairs linked to the selected recalls from sales of the corporation's remaining products. Comparisons within ATC3 and ATC4 classes examine potential compensation within the affected therapeutic market, while the firm-wide comparison also allows for offsets through products sold in other therapeutic classes. Thus, these descriptive comparisons help assess how revenue losses are distributed across the recalling firm's portfolio, and complement the class-level first-stage analysis, which examines total market revenues, including those earned by competing firms.

Table E.1 asks whether higher recall intensity last year is associated with lower sales in one part of a firm's portfolio and higher sales elsewhere. Panel A follows a fixed set of products linked to the selected recalls during the study period. Panel B covers the firm's remaining products. Finally, Panel C sums their sales. The columns examine either the whole firm or its products within an ATC3 or ATC4 class. The idea is simply that lower sales in A accompanied by higher sales in B would suggest that other products offset at least part of the loss

Table E.1: Selected recalls and within-firm portfolio sales, with pre-recall controls

	Firm	Firm–ATC3	Firm–ATC4
	Annual sales (USD millions)		
Panel A: Identified product portfolio			
$\widetilde{recalls}_{t-1}$	−0.1062	−0.0604***	−0.0484***
	(0.2056)	(0.0110)	(0.0102)
Panel B: Remaining own-firm products			
$\widetilde{recalls}_{t-1}$	−2.9834	−0.0648	−0.0156
	(5.8578)	(0.2553)	(0.2302)
Panel C: Total sales			
$\widetilde{recalls}_{t-1}$	−3.0896	−0.1252	−0.0639
	(5.8704)	(0.2558)	(0.2307)
Product age and its square, $t-2$	Yes	Yes	Yes
Log own product count, $t-2$	Yes	Yes	Yes
Generic product share, $t-2$	Yes	Yes	Yes
Unit fixed effects	Yes	Yes	Yes
Year dummies	Yes	Yes	Yes
Observations	5571	37914	42287
Groups	1076	7555	8471
Corporation clusters	.	.	.

Notes: Standard errors in parentheses, clustered by corporation. $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$. Each cell reports the coefficient on previous-year selected issue families per 100 own marketed products. Identified products are a fixed portfolio linked to specification 2 over 2006–2012; remaining products are all other products of the same corporation within the column's aggregation. The firm column pools the ATC3 benchmark's selected products across classes; the ATC4 column retains its own source support. Within each column, panels use identical observations, unit-year effects, and controls; therefore, the coefficients in A and B sum to C. A positive B alongside a negative A would indicate an offsetting association. A statistically insignificant C does not establish complete compensation. Outcome years are 2007–2013; units must have positive sales at $t-2$, before the lagged recall, but may have zero current sales. Controls are measured at $t-2$, before the lagged recall: the mean age of own marketed products and its square, log product count and the fraction with any generic-type presentation. Firm age is absorbed by the fixed effects. Earlier measurement does not establish control exogeneity. Sales retain the source price basis, with no CPI adjustment. These are descriptive portfolio regressions, distinct from the class-level IV first stages. The retrospective product grouping, other recalls, entry and ownership changes preclude a causal substitution interpretation.

Having Panel A only significant means that we have evidence that higher recall intensity is associated with lower sales in the identified portfolio. So our results are compatible with firms not being able to compensate, in the short run, the loss provoked by a recall with the sales of other products in the same ATC. But the results for the firm's other products are too uncertain to tell whether they gain sales and make up that loss, remain unchanged, or also lose sales. Their estimated coefficients are negative, so they do not point towards compensation, but they are too imprecise to rule it out.

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